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Progress Report

2026-10-01

Modeling DUR Durations

A model was built to predict DUR durations for the following reasons:

- The analysis in [Would Grouping Similar Work Make Pharmacists Faster? What the data says](#) in concludes that there is no effect if the previous task is the same. We would like to know if we can control for other factors than MC3, Cohort or part number, the result would change? And is there a definition of similarity that is more suitable to use. Can we get bounds on the uncertainty?

Consider the 2 task sequences of the same user:

MC3	SEQUENCE ORDER	PART_NUMBER	PET	DURATION	START TIME	UPDATED FIELDS	APPROVAL CHANNEL	INITIATION CHANNEL
Parasiticide	1	146066	Dog	40	2026-04-30 06:42:54.000 +0000		PH	AUTOSHIP
Parasiticide	2	152702	Dog	27	2026-04-30 06:43:36.000 +0000	directions	PH	AUTOSHIP
Parasiticide	3	146391	Dog	40	2026-04-30 06:44:06.000 +0000		PH	AUTOSHIP
Parasiticide	4	151583	Dog	7	2026-04-30 06:44:48.000 +0000		PH	AUTOSHIP
Parasiticide	5	146138	Dog	7	2026-04-30 06:44:56.000 +0000		PH	AUTOSHIP
Parasiticide	6	146142	Dog	43	2026-04-30 06:45:05.000 +0000	directions	PH	AUTOSHIP
Parasiticide	7	146142	Dog	35	2026-04-30 06:45:49.000 +0000	directions	PH	AUTOSHIP
Parasiticide	8	146064	Dog	39	2026-04-30 06:46:26.000 +0000	directions	PH	AUTOSHIP
Parasiticide	9	146167	Dog	12	2026-04-30 06:47:07.000 +0000		PH	AUTOSHIP
Parasiticide	10	146322	Dog	9	2026-04-30 06:47:21.000 +0000	directions	PH	AUTOSHIP
Parasiticide	11	146167	Dog	97	2026-04-30 06:47:32.000 +0000	directions	PH	AUTOSHIP
Parasiticide	12	146050	Dog	6	2026-04-30 06:49:11.000 +0000		PH	AUTOSHIP
Parasiticide	13	146387	Dog	54	2026-04-30 06:49:19.000 +0000		PH	AUTOSHIP

and

MC3	SEQUENCE ORDER	PART_NUMBER	PET	DURATION	START TIME	UPDATED FIELDS	APPROVAL CHANNEL	INITIATION CHANNEL
Parasiticide	1	158995	Cat	11	2026-07-15 02:17:23.000 +0000		PH	OMS
Parasiticide	2	158995	Cat	35	2026-07-15 02:17:36.000 +0000		PH	OMS
Parasiticide	3	158995	Cat	3	2026-07-15 02:18:13.000 +0000		PH	OMS
Parasiticide	4	158995	Cat	13	2026-07-15 02:18:17.000 +0000		PH	OMS
Parasiticide	5	158995	Cat	13	2026-07-15 02:18:31.000 +0000		PH	OMS
Parasiticide	6	158995	Cat	6	2026-07-15 02:18:46.000 +0000		PH	OMS
Parasiticide	7	158995	Cat	16	2026-07-15 02:18:53.000 +0000		PH	OMS
Parasiticide	8	158995	Cat	13	2026-07-15 02:19:11.000 +0000		PH	OMS
Parasiticide	9	158995	Cat	6	2026-07-15 02:19:25.000 +0000		PH	OMS
Parasiticide	10	158995	Cat	50	2026-07-15 02:19:32.000 +0000		PH	OMS
Parasiticide	11	158998	Cat	11	2026-07-15 02:20:23.000 +0000		PH	OMS

MC3	SEQUENCE ORDER	PART_NUMBER	PET	DURATION	START TIME	UPDATED FIELDS	APPROVAL CHANNEL	INITIATION CHANNEL
Parasiticide	12	158998	Cat	29	2026-07-15 02:20:36.000 +0000		PH	OMS

There seems to be an inherent variability in the DUR times looking at the second trace. Using a model, we would like to quantify this.

- Another motivation to simulate what-if scenarios where different people do a task and see statistically how the DUR duration would change. We can use the model to predict DUR times for task - user combinations that did not occur in the past with possible error bounds.

The model does predict that if the previous task is "similar" for a user, there is a statistically significant effect. However, the model also predicts that there is a lot of noise.

The predicted effect on various DUR tasks is:

Duration without same preceding task	Duration with same preceding task	-1 SD bound	+1 SD bound	Difference	Percentage change
15.00 s	14.16 s	6.84 s	30.70 s	-0.84 s	-5.58%
50.00 s	46.93 s	21.09 s	110.44 s	-3.07 s	-6.14%
150.00 s	139.96 s	58.53 s	357.84 s	-10.04 s	-6.69%

The model predicts that users differ in their processing speed, products differ in speed and the day of the week does not matter.

Obviously, if there is an alteration of prescription data, the model predicts longer DUR times.

The question is how reliable will our estimates be for any what if scenarios regarding sequencing?

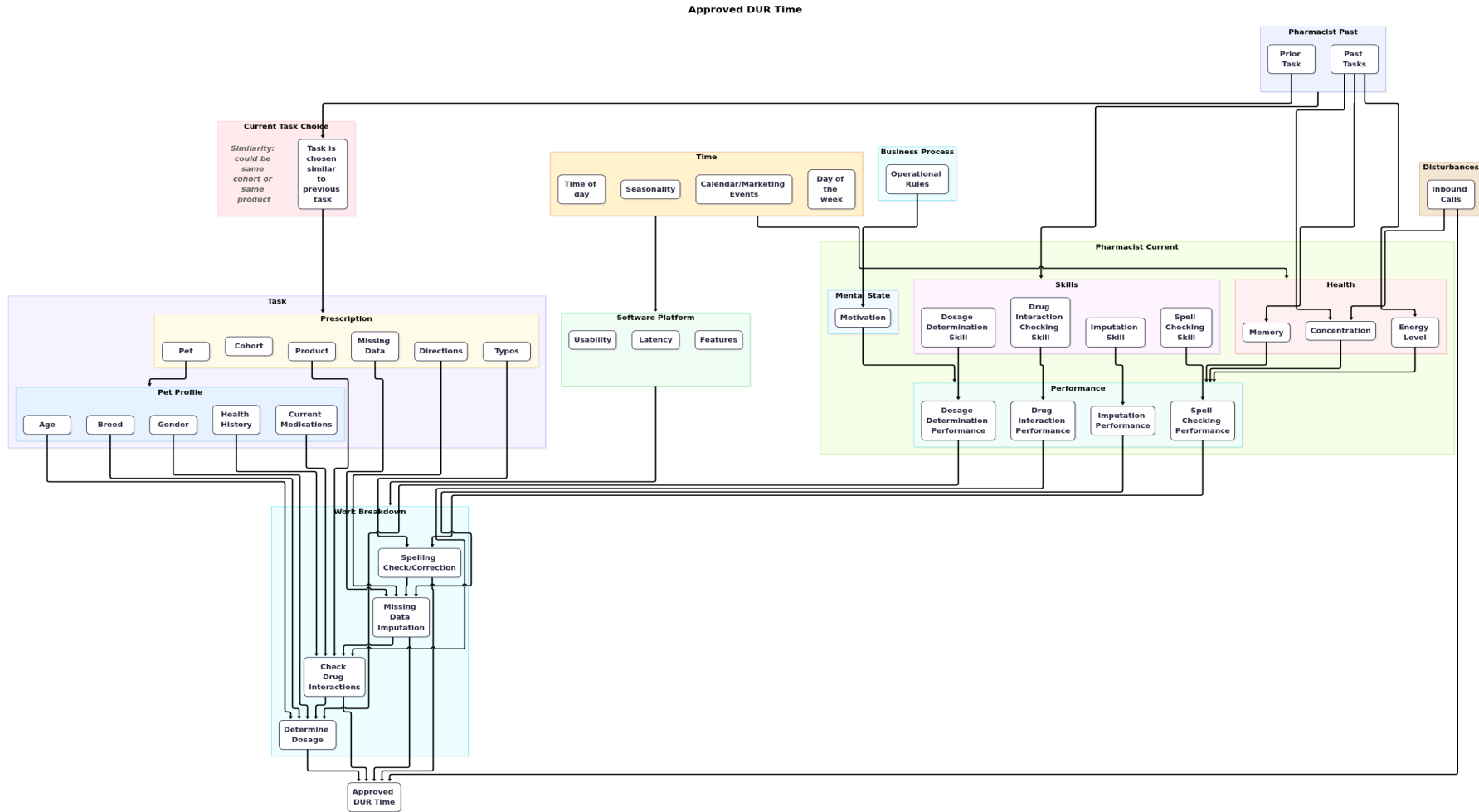
EDA on Factors Affecting DUR

See [Factors Affecting DUR](#) in the Appendix.

We assume DUR durations are affected by these for main groups of factors

1. The user
 - Are some users inherently slower? Are they slower or faster all across the board.
2. The prescription
 - Medication
 - Content (typos, missing information)
 - Pet
 - Pet health history
3. Business Process
4. Outside disturbances such as inbound calls
5. The user's task history
 - Did they do similar tasks before
6. Environment
7. Software Platform.

A more detailed breakdown can be found [here](#).



For details see [Model Details](#) in the Appendix.

Appendix

Model Details

Final MC3/COH/PETTYPE mixed-effects model The model is built to understand the factors affecting approved DUR task durations. It uses MC3, cohort, and pet type to define groups of similar DUR tasks.

The model controls for weekday, approval channel, prescription source, parent part number, and user (pharmacist, pharmacy technician, etc.) and weekday and alterations to the data e.g. update the directions on the prescription.

There are two treatments based on whether the current task is adjacent to another task in the same MC3, cohort, and pet type equivalence class for a given user. Adjacent tasks must be no more than one hour apart. Failed DURs and DUR tasks that overlap in time with other DUR tasks are excluded; these cases occur for

certain orders. If a preceding task belongs to the same equivalence class, the **SAME_PRECEDING** treatment is true. If the following task belongs to the same equivalence class, the **SAME_FOLLOWING** treatment is true. **SAME_FOLLOWING** is expected to have a coefficient near zero. It exists to check whether the model captures confounding factors adequately.

The model assigns the difference of each user from a mean performance. The model does the same for parent part number and the calendar day.

The model predicts that if the preceding task is the same for a user, there is a decrease in the duration of a DUR. The model also suggests that given the same MC3, cohort, pet type and parent product type, there is no statistically significant effect from the future.

The week day is not significantly important, the approval channel and prescription source matter. Customer originating prescriptions take longer.

Purpose and analysis sample

This is the final model. It estimates how same-cell sequence position relates to DUR duration while controlling for correction status, weekday, approval channel, prescription source, and MC3/COH/PETTYPE cell. It includes crossed random intercepts for user, process date, and parent part number.

The source is EDLDB_DEV.PET_HEALTH_ANALYTICS_SANDBOX.MY_RXP_VALID_DUR_TASKS. Eligible rows have non-null user, task, process-start time, MC3_COH_PETTYPE_SEQN, and MC3_COH_PETTYPE_BATCH_LEN; sequence and batch length are at least one, and sequence does not exceed batch length. Duration is IMPUTED_DWELL_IN_PROGRESS_TO_CLOSE.

Let Y_i be the positive imputed duration in seconds for task i . One global cutoff is calculated over all otherwise eligible tasks with Snowflake's APPROX_PERCENTILE(Y , 0.95) function call:

$$q_{0.95} = Q_{0.95}(Y).$$

The fitted sample keeps tasks satisfying $0 < Y_i < q_{0.95}$; the upper bound is strict. MC3, COH, PETTYPE, parent part number, approval channel, and prescription source blanks are represented by <Missing>.

Derived indicators

For sequence number S_i , batch length L_i , and correction count N_i , the three binary fixed effects are

$$P_i = \mathbf{1}(L_i > 1 \wedge S_i > 1), \quad F_i = \mathbf{1}(L_i > 1 \wedge S_i < L_i), \quad C_i = \mathbf{1}(N_i > 0).$$

Thus P_i is **SAME_PRECEDING**, F_i is **SAME_FOLLOWING**, and C_i is **HAS_CORRECTION**.

Outcome transformation

The retained durations receive one global Box-Cox transformation:

$$Z_i = g_\lambda(Y_i) = \begin{cases} \frac{Y_i^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log(Y_i), & \lambda = 0. \end{cases}$$

The maximum-likelihood transformation estimate is $\hat{\lambda} = -0.08159843$. All coefficients, random effects, standard deviations, residuals, RMSE, and diagnostics are therefore on the Box-Cox scale.

Complete model formula

Index task, fixed-effect cell, user, date, and parent part by i, c, u, d, p . The model is

$$\begin{aligned}
Z_i = & \gamma_{c(i)} + \beta_P P_i + \beta_F F_i + \beta_C C_i \\
& + \sum_{w \neq \text{Sun}} \delta_w \mathbf{1}(W_i = w) \\
& + \sum_{a \neq a_0} \theta_a \mathbf{1}(A_i = a) + \sum_{s \neq s_0} \phi_s \mathbf{1}(R_i = s) \\
& + b_{u(i)} + h_{d(i)} + r_{p(i)} + \epsilon_i.
\end{aligned}$$

Here $\gamma_{c(i)}$ is a fixed effect shared by all tasks in the same MC3 + COH + PETTYPE cell. The model uses a full set of cell effects and no separate overall intercept. Sunday is the weekday reference. The data-dependent categorical references are:

- approval channel a_0 : PH
- prescription source s_0 : PRACTICE_HUB

Each reference is the most frequent retained level; lexicographic order breaks a frequency tie. The crossed random terms and level-1 error are mutually independent and satisfy

$$b_u \sim N(0, \sigma_u^2), \quad h_d \sim N(0, \sigma_d^2), \quad r_p \sim N(0, \sigma_p^2), \quad \epsilon_i \sim N(0, \sigma_e^2).$$

The fixed effects and variance components are estimated jointly by REML. The conditional fitted value and residual are

$$\widehat{Z}_i = \hat{\gamma}_{c(i)} + x_i^T \hat{\beta} + \hat{b}_{u(i)} + \hat{h}_{d(i)} + \hat{r}_{p(i)}, \quad e_i = Z_i - \widehat{Z}_i.$$

Inference and fit statistics

For fixed effect k , the report uses

$$z_k = \frac{\hat{\beta}_k}{\text{SE}(\hat{\beta}_k)}, \quad p_k = 2\Phi(-|z_k|), \quad \text{CI}_{95} = \hat{\beta}_k \pm 1.96 \text{SE}(\hat{\beta}_k).$$

The fixed-effect table labels an estimate **Significant = Yes** when its 95% confidence interval excludes zero; intervals that cross or touch zero are marked **No**.

Random-effect SD intervals use an observed-information Wald approximation on the log-SD scale. The residual-SD interval uses the chi-square distribution. The reported conditional metrics include all fixed and random effects:

$$R_c^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (Z_i - \bar{Z})^2}, \quad \text{RMSE}_c = \sqrt{\frac{1}{n} \sum_i e_i^2}.$$

Sample and fit

Quantity	Value
Tasks	3,433,218
MC3/COH/PETTYPE cells	265
Users	340

Quantity	Value
Dates	182
Parent part numbers	885
Approval-channel levels	9
Prescription-source levels	11
Global approximate P95 cutoff (seconds)	156.598673
Box-Cox lambda	-0.08159843
Conditional R-squared	0.308784
Conditional RMSE (Box-Cox scale)	0.603674
REML negative log-likelihood	3142827.127402
Converged	True

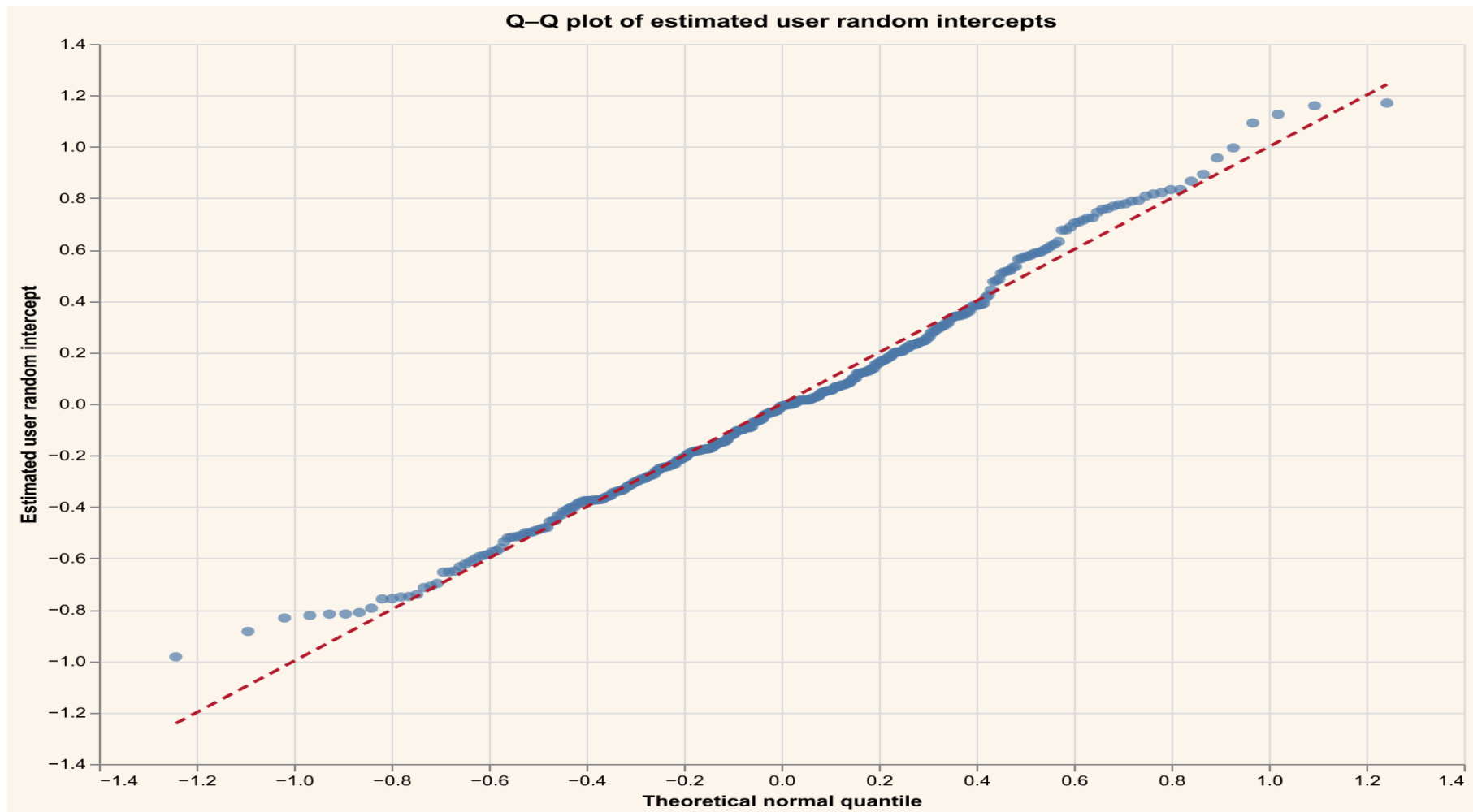
Fixed effects

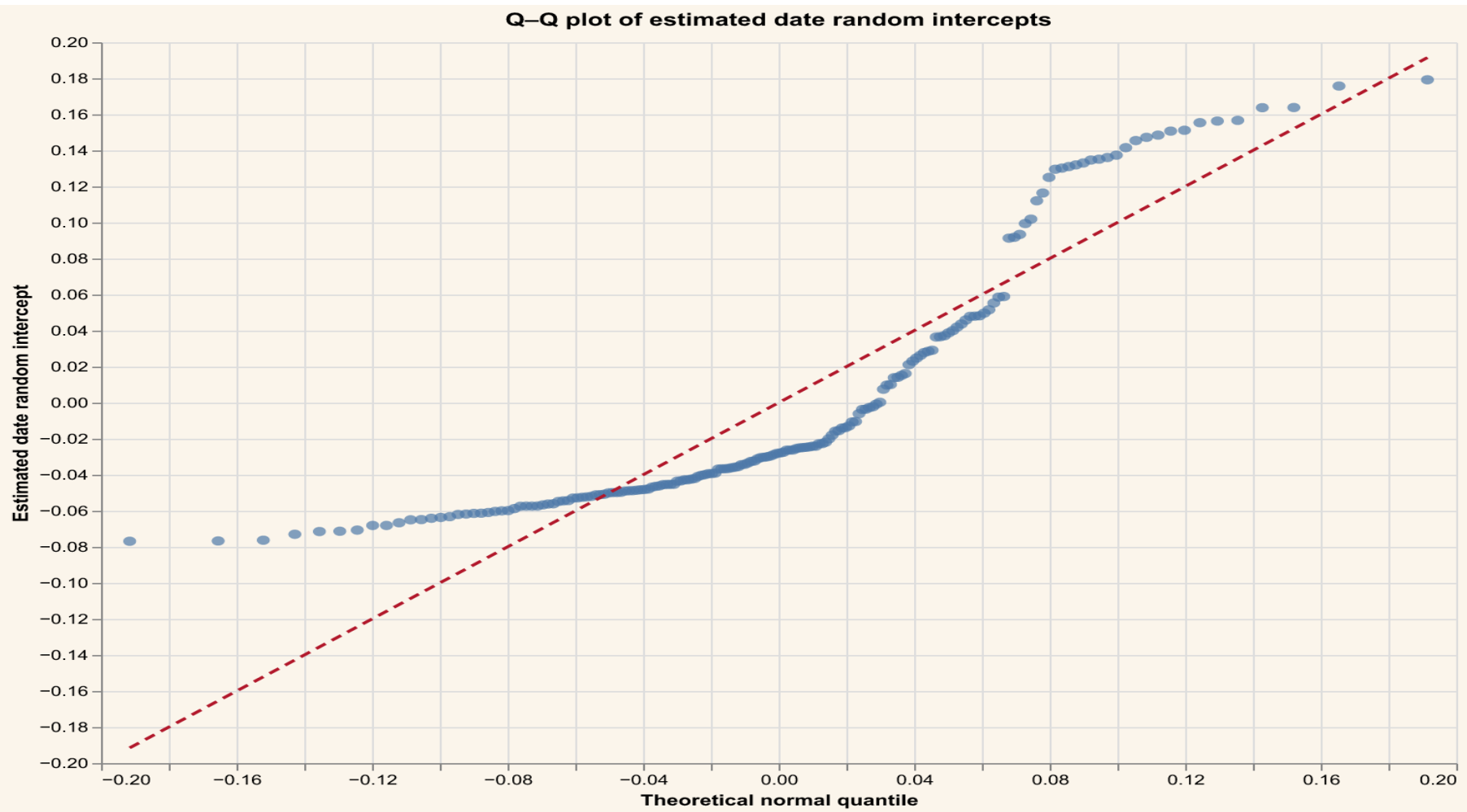
Term	Estimate	SE	z	p-value	95% CI	Significant
SAME_PRECEDING	-0.046157	0.000835	-55.2813	<1e-300	[-0.047793, -0.044520]	Yes
SAME_FOLLOWING	0.001024	0.000835	1.2259	0.220231	[-0.000613, 0.002661]	No
HAS_CORRECTION	0.271146	0.000788	344.0672	<1e-300	[0.269601, 0.272690]	Yes
WEEKDAY_MON	0.006416	0.019649	0.3265	0.74404	[-0.032096, 0.044927]	No
WEEKDAY_TUE	0.014299	0.019646	0.7278	0.466718	[-0.024206, 0.052803]	No
WEEKDAY_WED	0.010462	0.019650	0.5324	0.59444	[-0.028051, 0.048975]	No
WEEKDAY_THU	0.008138	0.019652	0.4141	0.678785	[-0.030379, 0.046655]	No
WEEKDAY_FRI	0.006015	0.019651	0.3061	0.75952	[-0.032499, 0.044530]	No
WEEKDAY_SAT	0.002131	0.019666	0.1084	0.913702	[-0.036413, 0.040675]	No
APPROVAL_CHANNEL[CLINIC_EMAIL vs PH]	0.235910	0.045211	5.2180	1.80853e-07	[0.147299, 0.324522]	Yes
APPROVAL_CHANNEL[CUSTOMER_MAIL vs PH]	0.472534	0.013061	36.1785	1.32431e-286	[0.446934, 0.498133]	Yes
APPROVAL_CHANNEL[DIGITAL vs PH]	0.146802	0.040724	3.6048	0.000312369	[0.066985, 0.226619]	Yes
APPROVAL_CHANNEL[FAX vs PH]	0.208015	0.011227	18.5276	1.23668e-76	[0.186010, 0.230020]	Yes
APPROVAL_CHANNEL[PARTNER vs PH]	-0.034275	0.007967	-4.3018	1.69381e-05	[-0.049891, -0.018659]	Yes
APPROVAL_CHANNEL[RHAPSODY vs PH]	0.075320	0.052573	1.4327	0.15195	[-0.027721, 0.178362]	No
APPROVAL_CHANNEL[VERBAL vs PH]	-0.017687	0.010665	-1.6585	0.097222	[-0.038589, 0.003215]	No
APPROVAL_CHANNEL[VET vs PH]	0.003570	0.105720	0.0338	0.973059	[-0.203638, 0.210779]	No
PRESCRIPTION_SOURCE[CLINIC_EMAIL vs PRACTICE_HUB]	-0.003227	0.011181	-0.2886	0.772878	[-0.025141, 0.018687]	No
PRESCRIPTION_SOURCE[CS_FIRST_CONTACT vs PRACTICE_HUB]	-0.086051	0.203897	-0.4220	0.673003	[-0.485682, 0.313580]	No
PRESCRIPTION_SOURCE[CUSTOMER_MAIL vs PRACTICE_HUB]	0.044423	0.012813	3.4672	0.000525987	[0.019311, 0.069536]	Yes
PRESCRIPTION_SOURCE[CVC vs PRACTICE_HUB]	0.023167	0.052511	0.4412	0.659085	[-0.079753, 0.126087]	No
PRESCRIPTION_SOURCE[DIGITAL vs PRACTICE_HUB]	0.020414	0.040106	0.5090	0.610753	[-0.058192, 0.099020]	No
PRESCRIPTION_SOURCE[FAX vs PRACTICE_HUB]	-0.004739	0.011224	-0.4222	0.67286	[-0.026738, 0.017260]	No
PRESCRIPTION_SOURCE[PARTNER_API vs PRACTICE_HUB]	0.018961	0.007944	2.3869	0.0169906	[0.003392, 0.034530]	Yes
PRESCRIPTION_SOURCE[RHAPSODY vs PRACTICE_HUB]	-0.029700	0.062733	-0.4734	0.6359	[-0.152655, 0.093254]	No
PRESCRIPTION_SOURCE[TELEMEDICINE vs PRACTICE_HUB]	0.185396	0.052942	3.5018	0.000462051	[0.081631, 0.289161]	Yes
PRESCRIPTION_SOURCE[VERBAL vs PRACTICE_HUB]	-0.002803	0.011736	-0.2388	0.81126	[-0.025805, 0.020200]	No

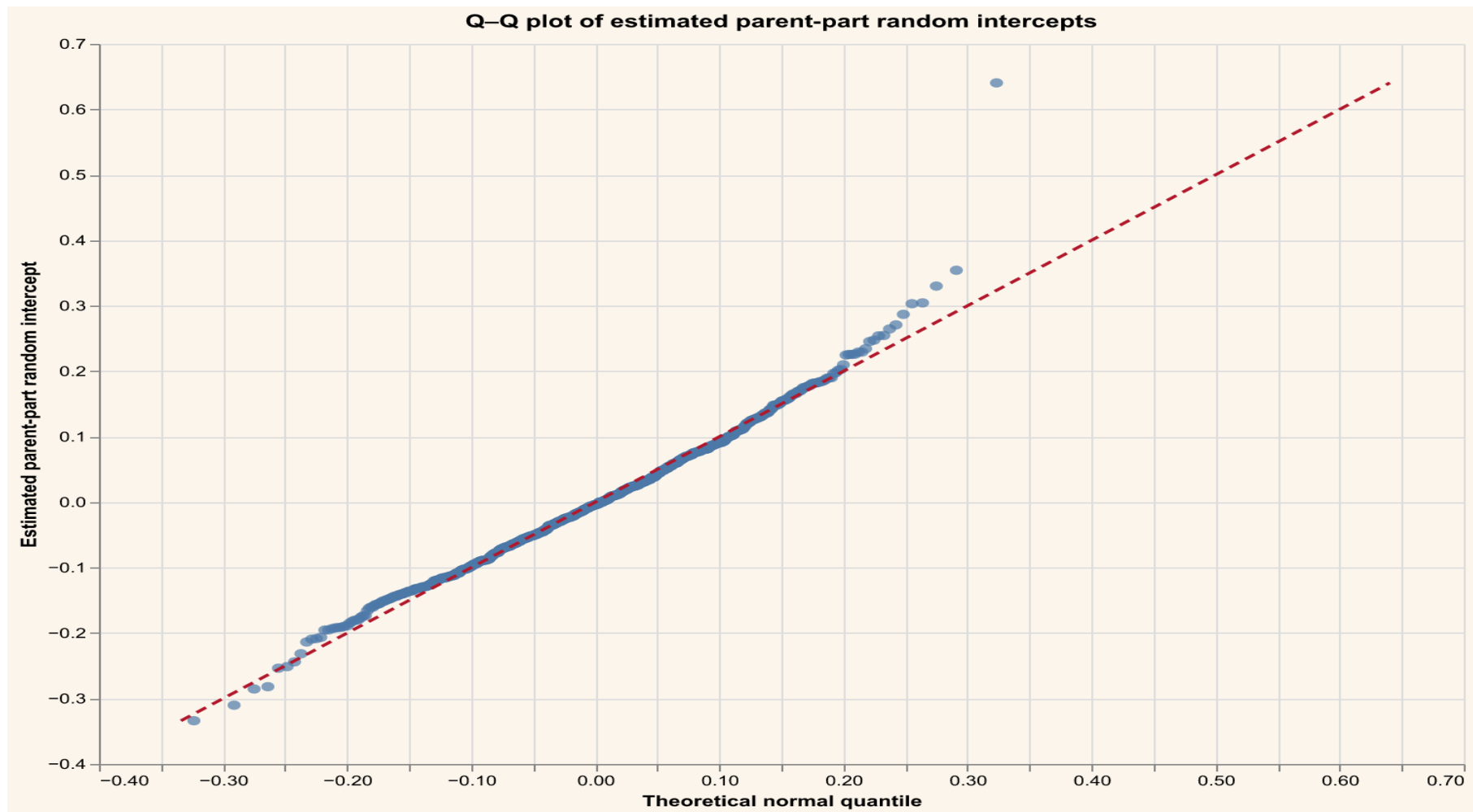
Variance components

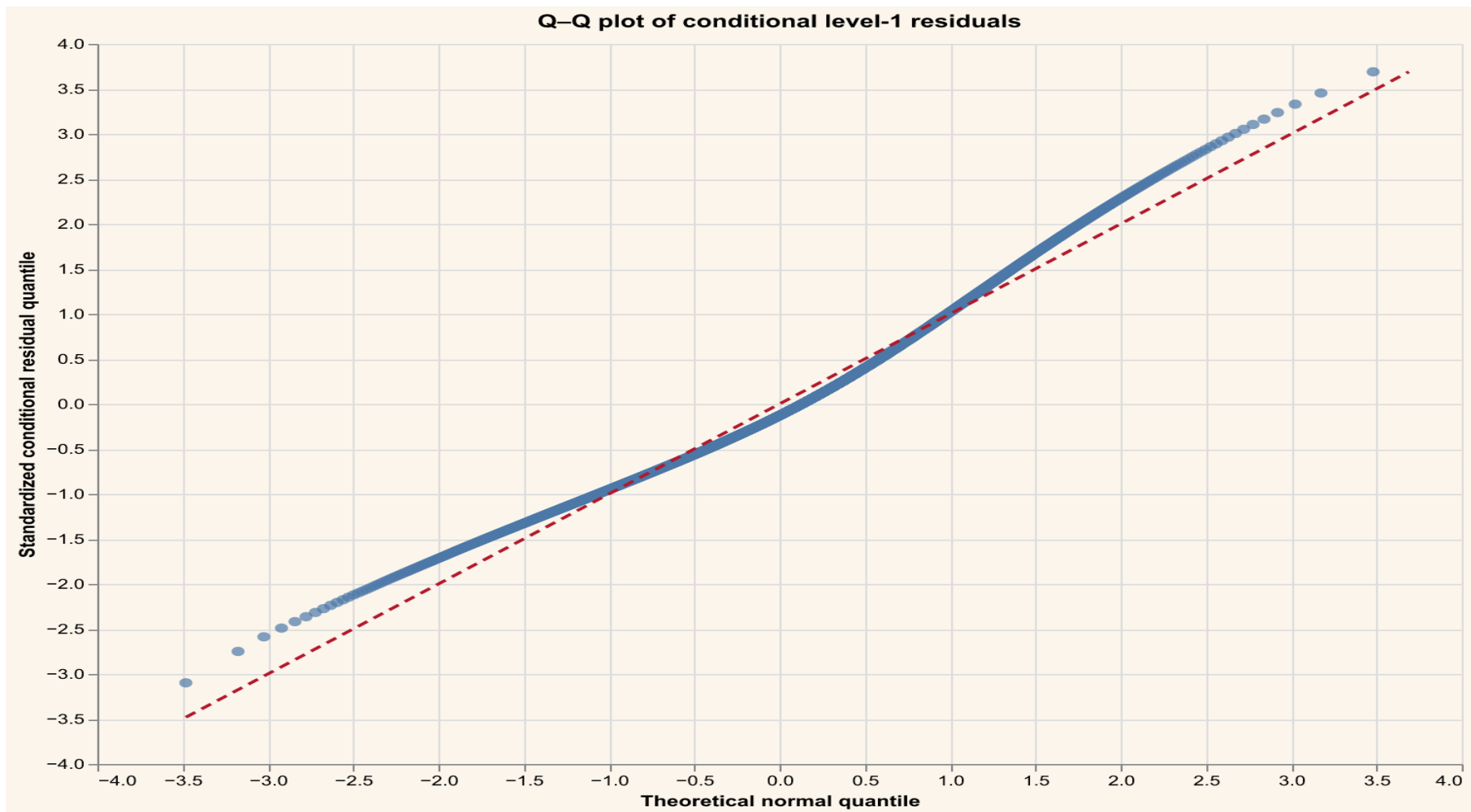
Component	SD	Approximate 95% CI
User random intercept	0.422318	[0.390500, 0.456728]
Date random intercept	0.070610	[0.063513, 0.078500]
Parent-part random intercept	0.114187	[0.107719, 0.121045]
Residual	0.603803	[0.603351, 0.604255]

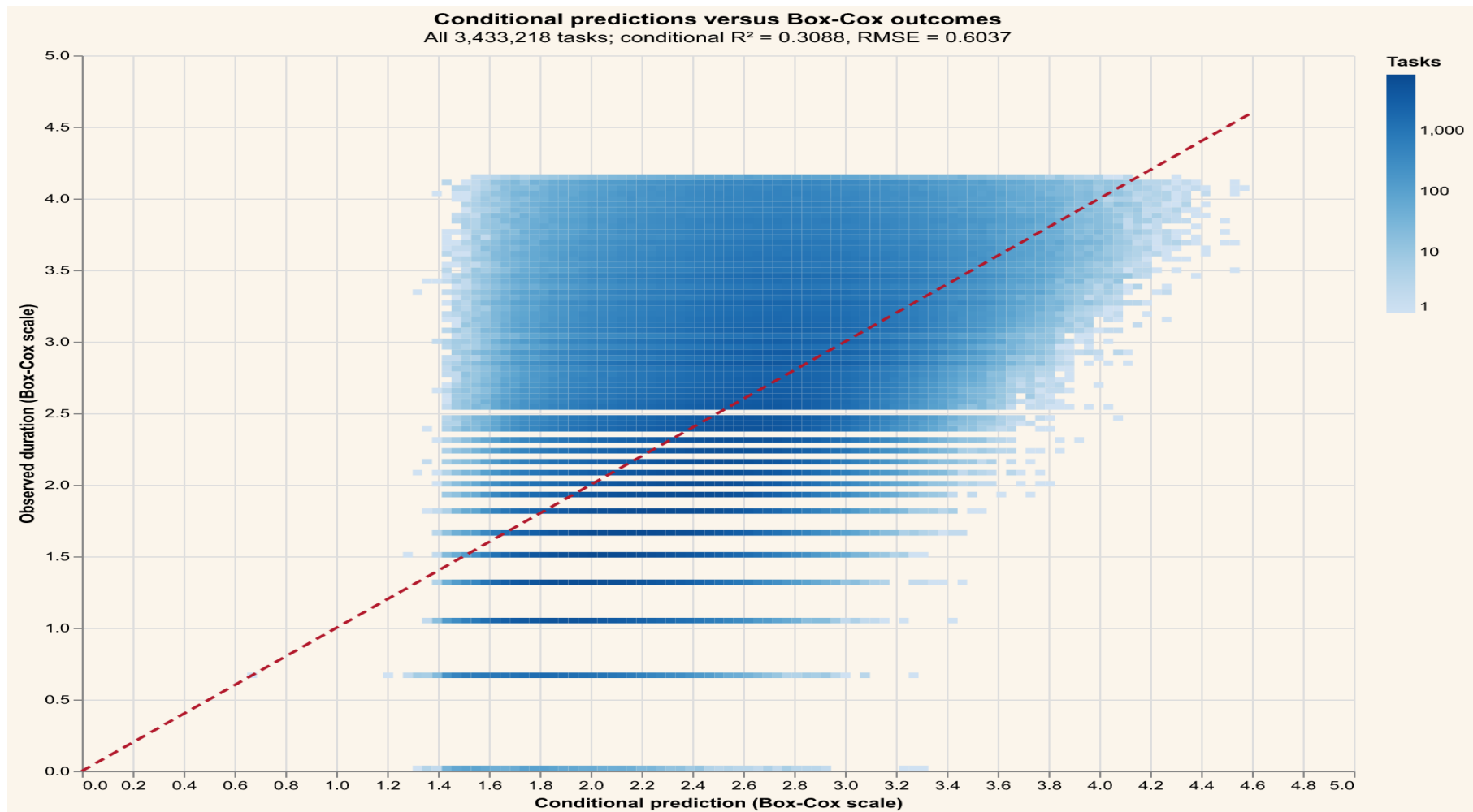
Diagnostics







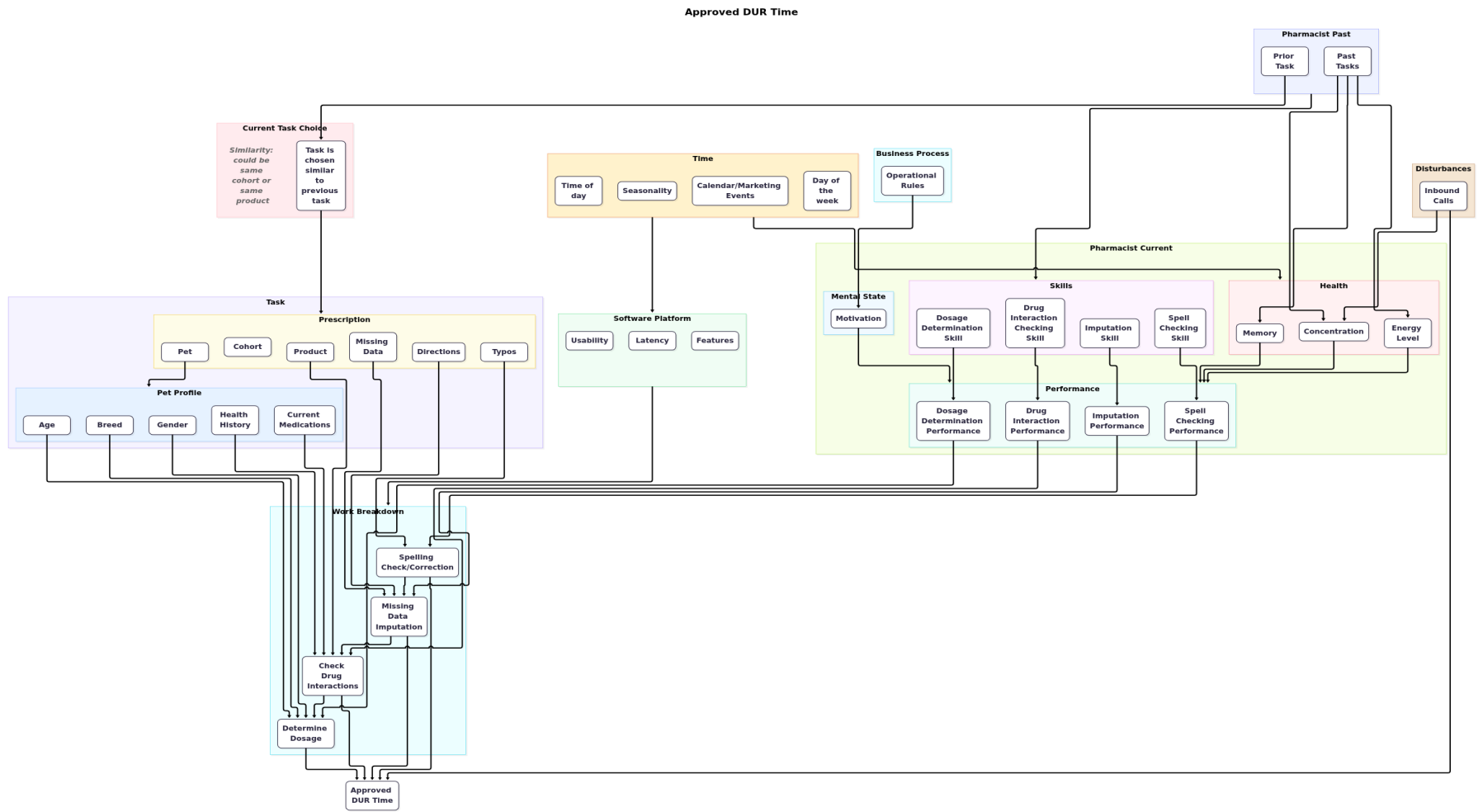




Artifacts

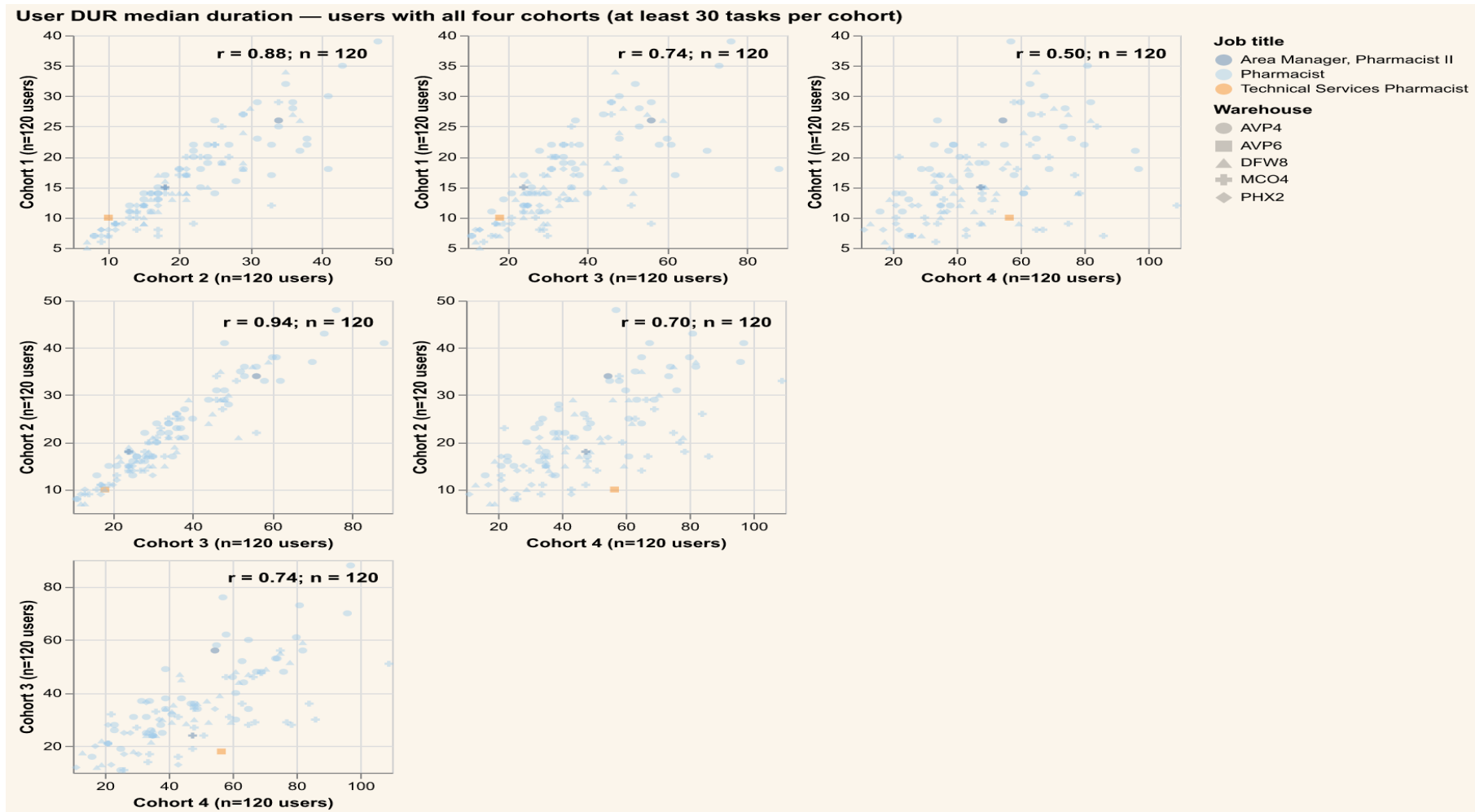
- [Generated SQL](#)
- [Raw Snowflake data](#)
- [Transformed model data](#)
- [Fixed-effect table](#)
- [User random effects](#)
- [Date random effects](#)
- [Parent-part random effects](#)

Factors Affecting Successful DURs



User Performance

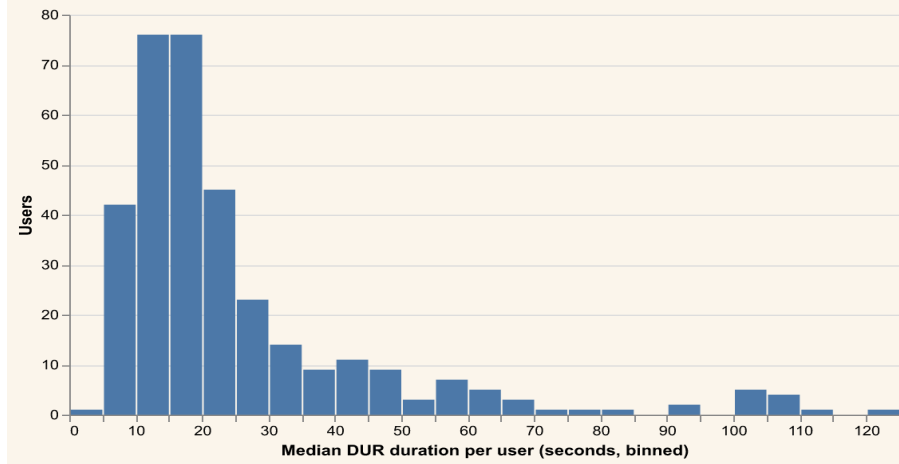
- Users' performance is correlated across all 4 cohorts.



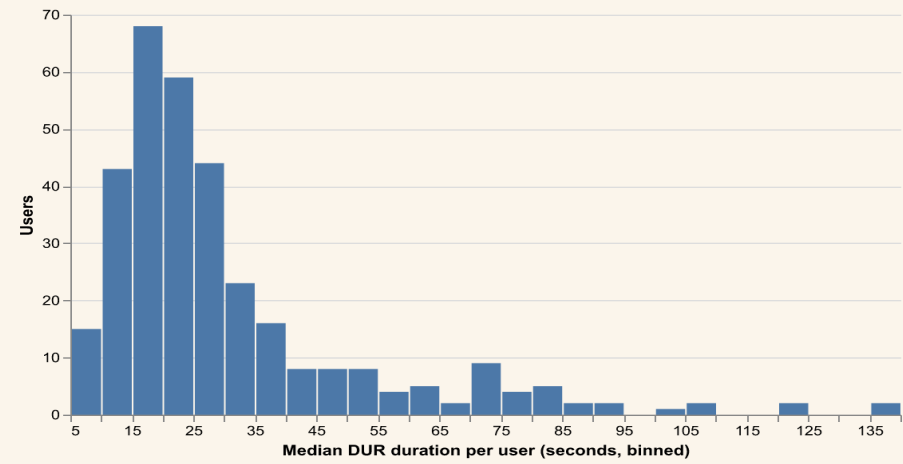
- Users' performance across cohorts vary

Distribution of median DUR duration per user

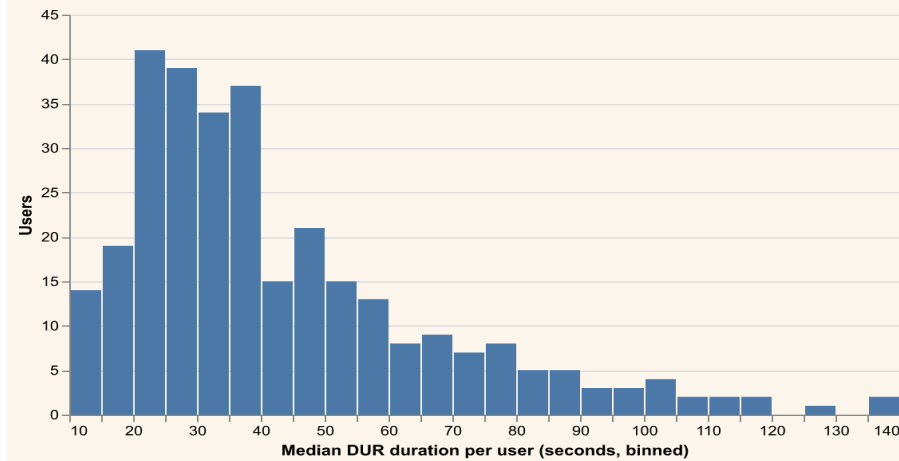
Cohort 1



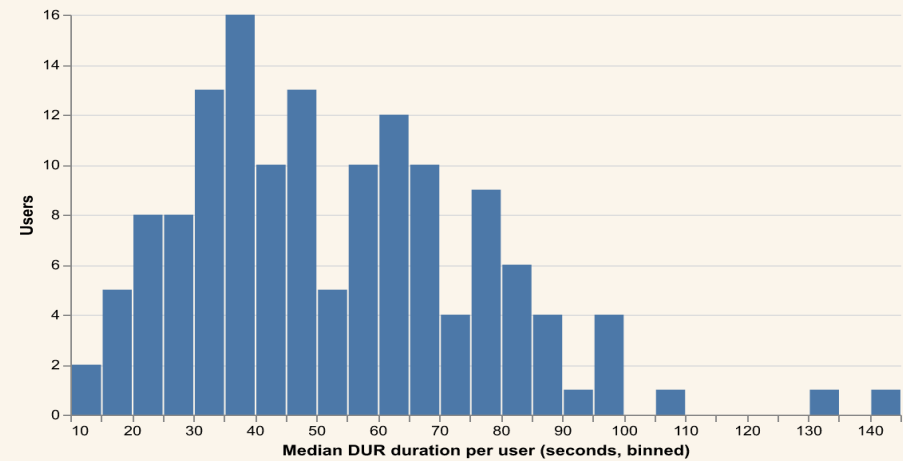
Cohort 2



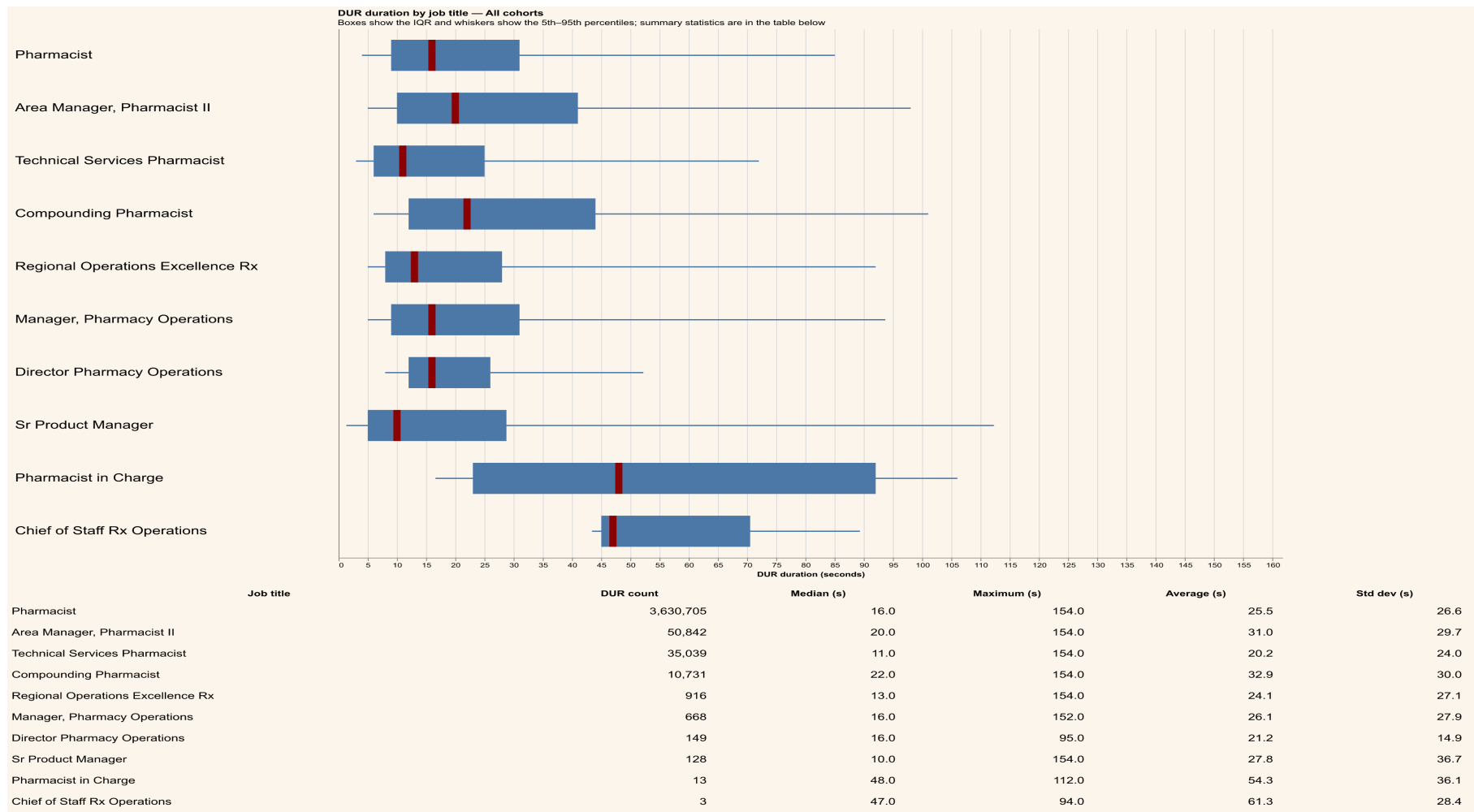
Cohort 3



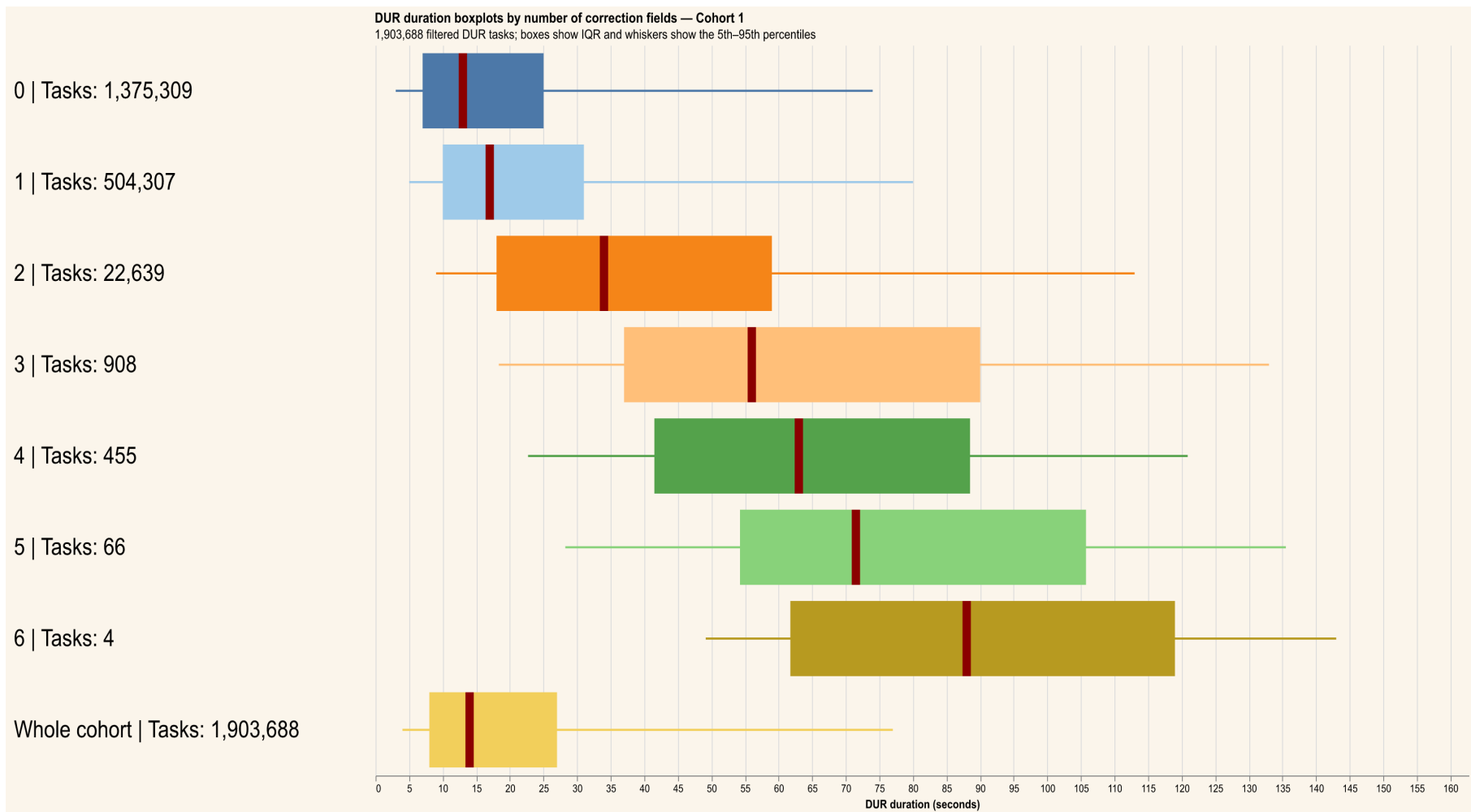
Cohort 4



- Effect of Job Title



Edits on Prescriptions The more the edits, the longer the DUR



DUR duration boxplots by number of correction fields — Cohort 2
 1,330,909 filtered DUR tasks; boxes show IQR and whiskers show the 5th–95th percentiles

0 | Tasks: 959,617

1 | Tasks: 351,511

2 | Tasks: 18,394

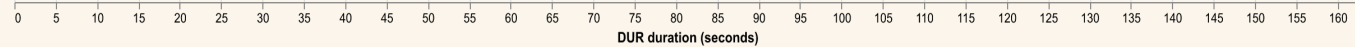
3 | Tasks: 1,080

4 | Tasks: 255

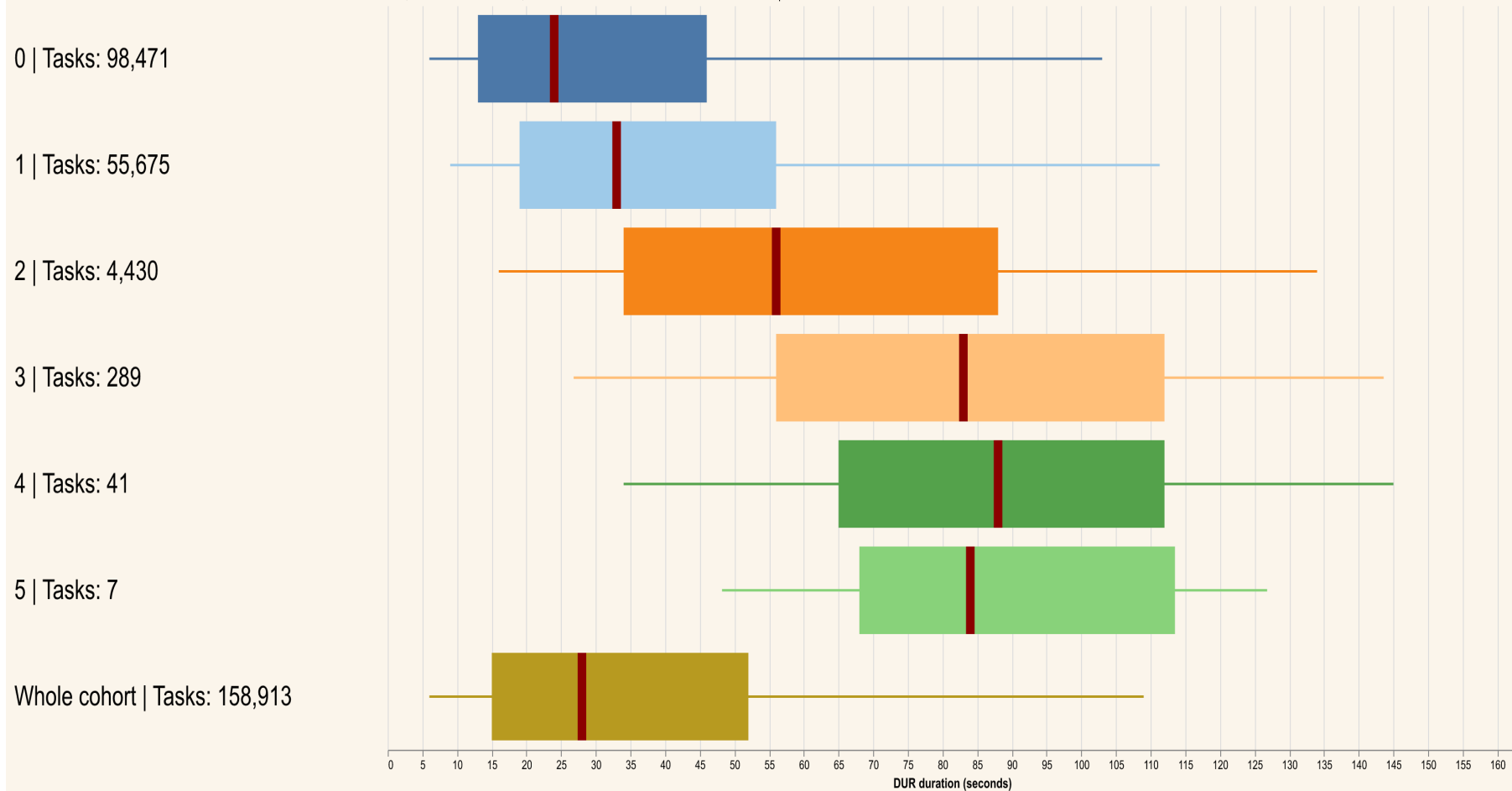
5 | Tasks: 50

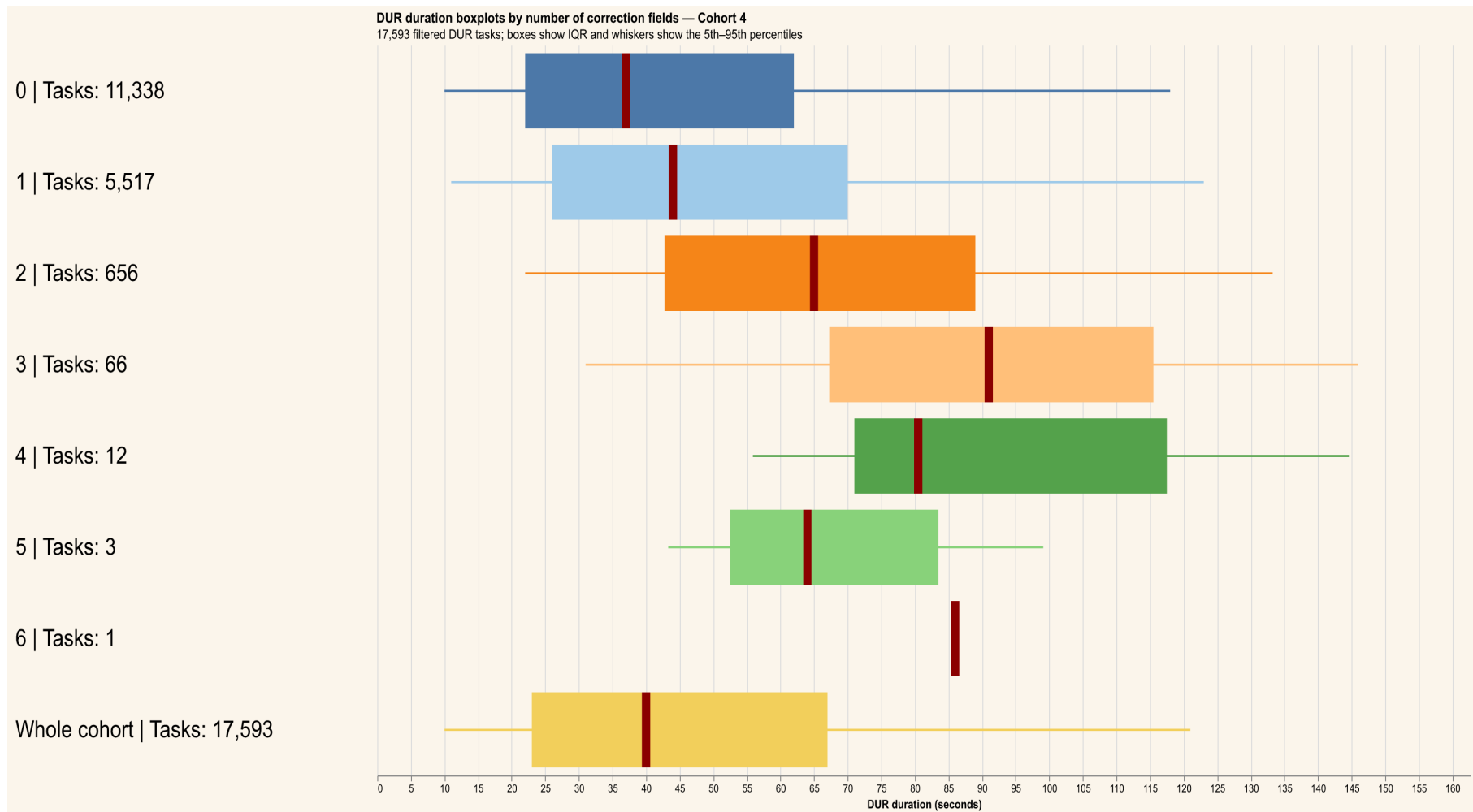
6 | Tasks: 2

Whole cohort | Tasks: 1,330,909

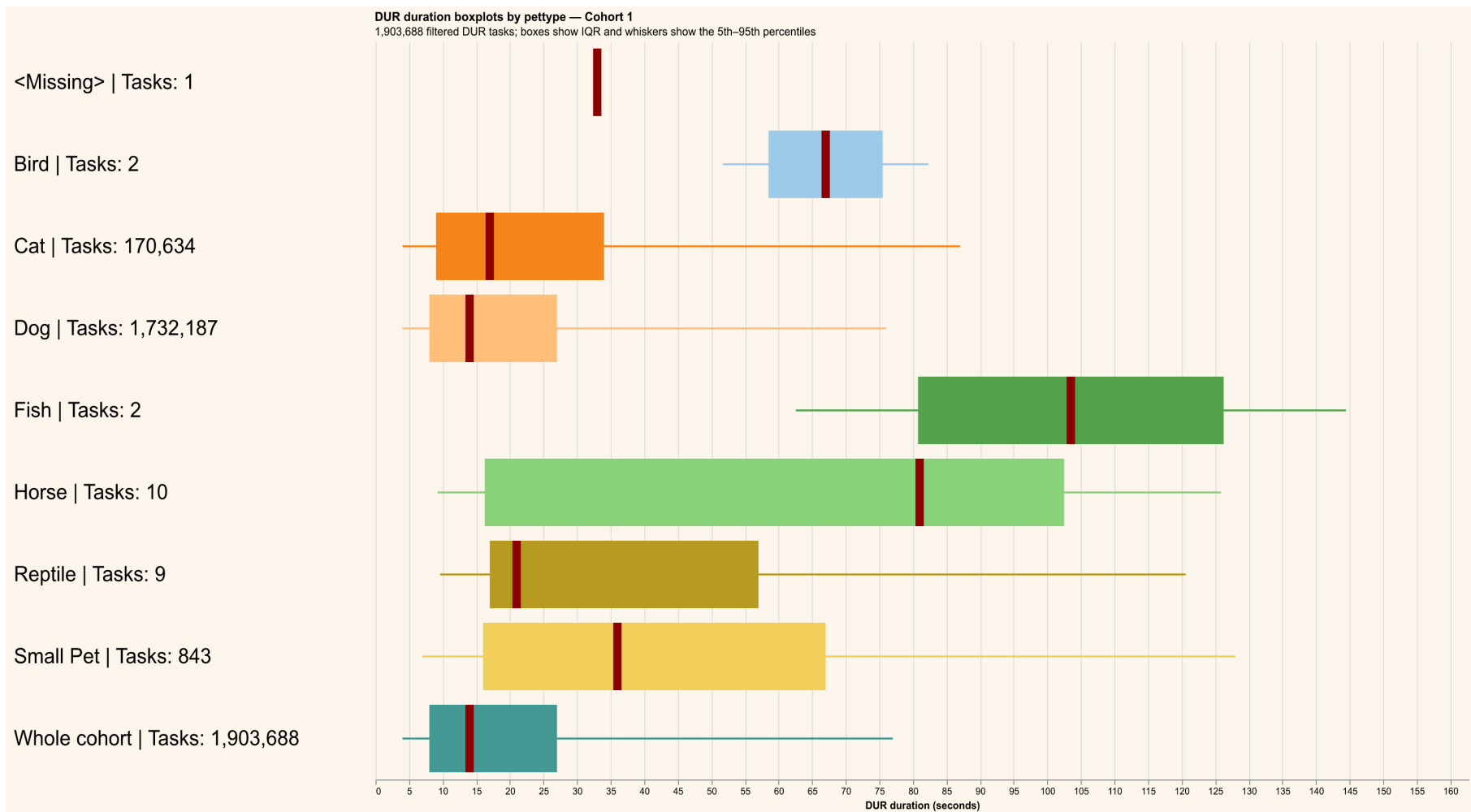


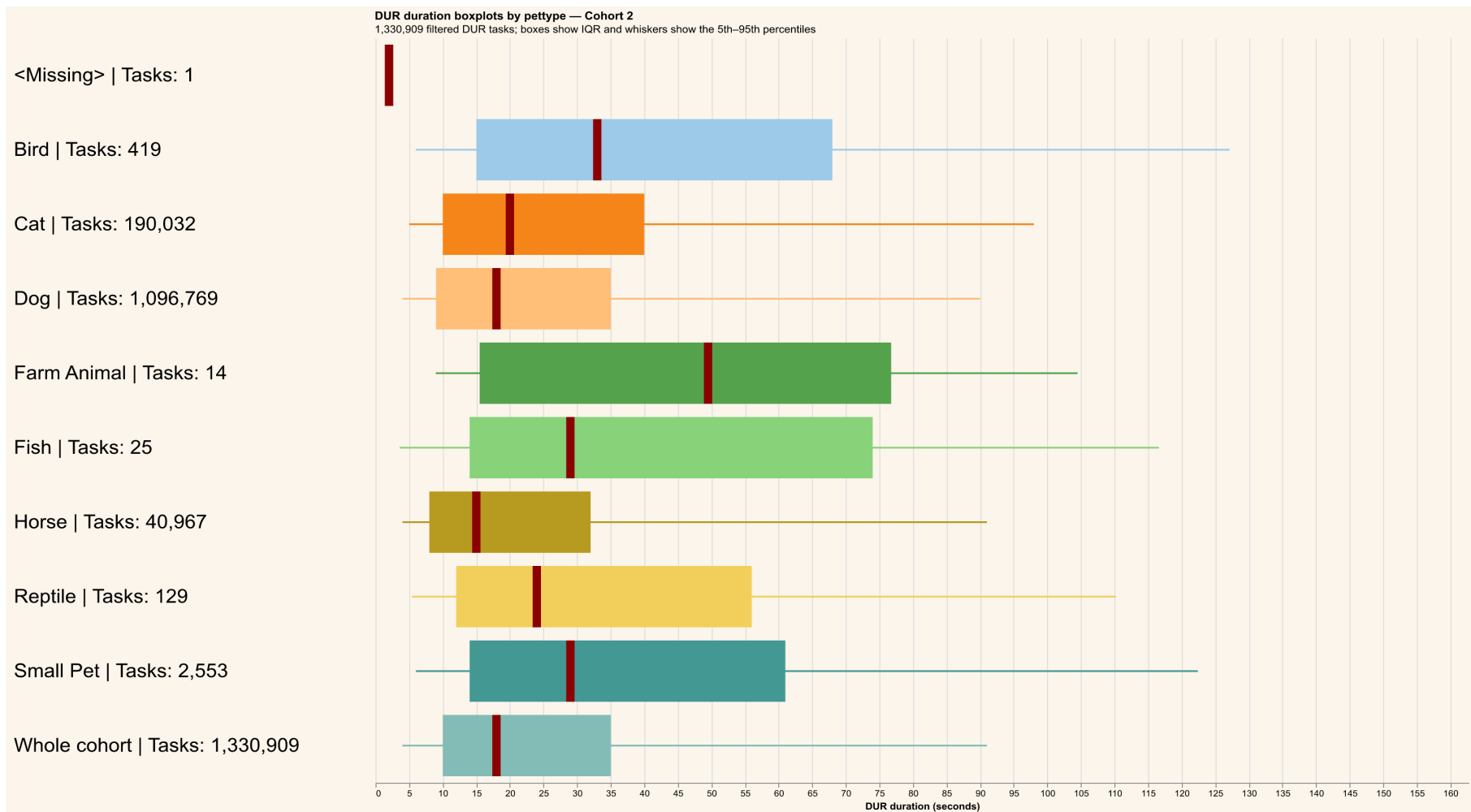
DUR duration boxplots by number of correction fields — Cohort 3
 158,913 filtered DUR tasks; boxes show IQR and whiskers show the 5th–95th percentiles

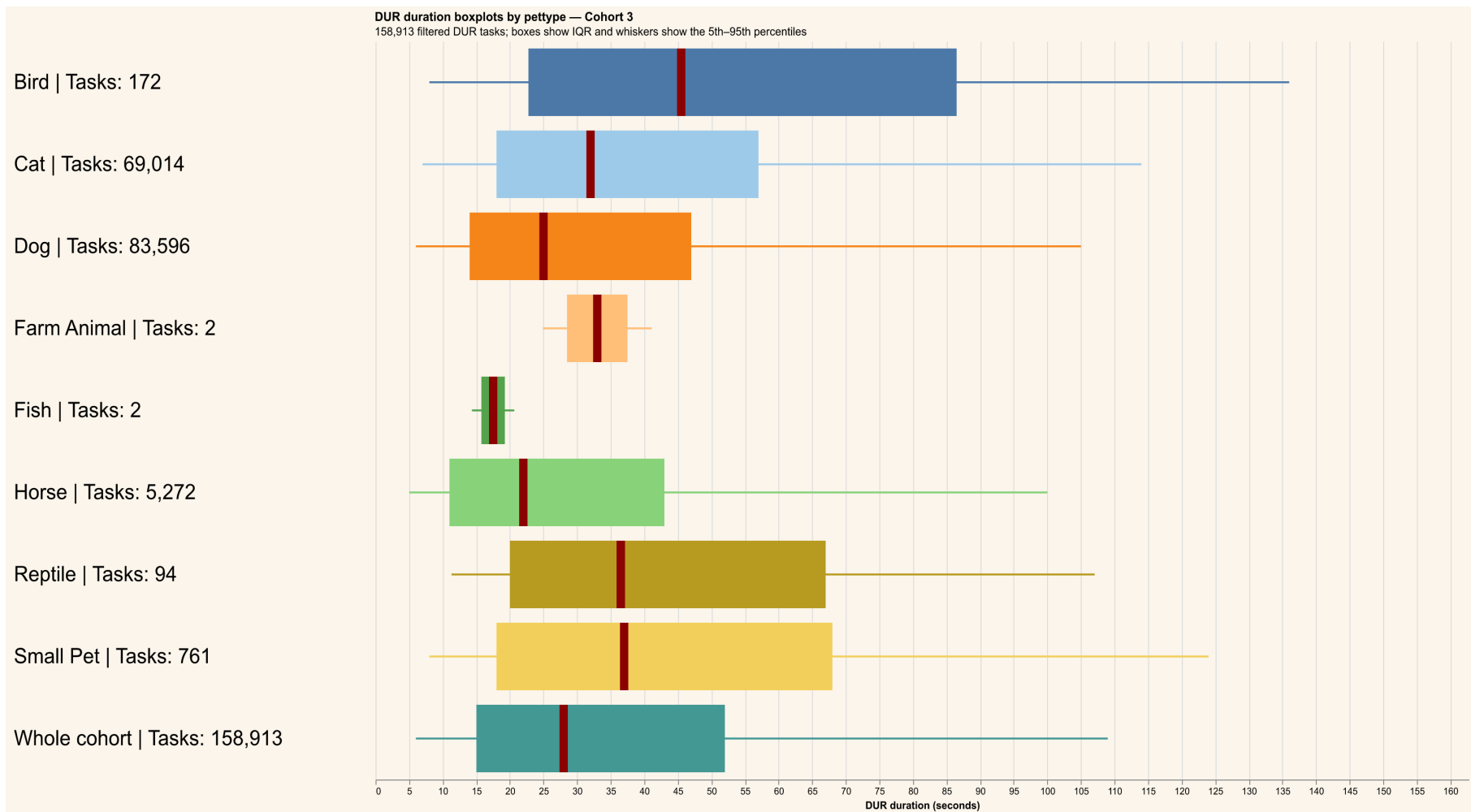


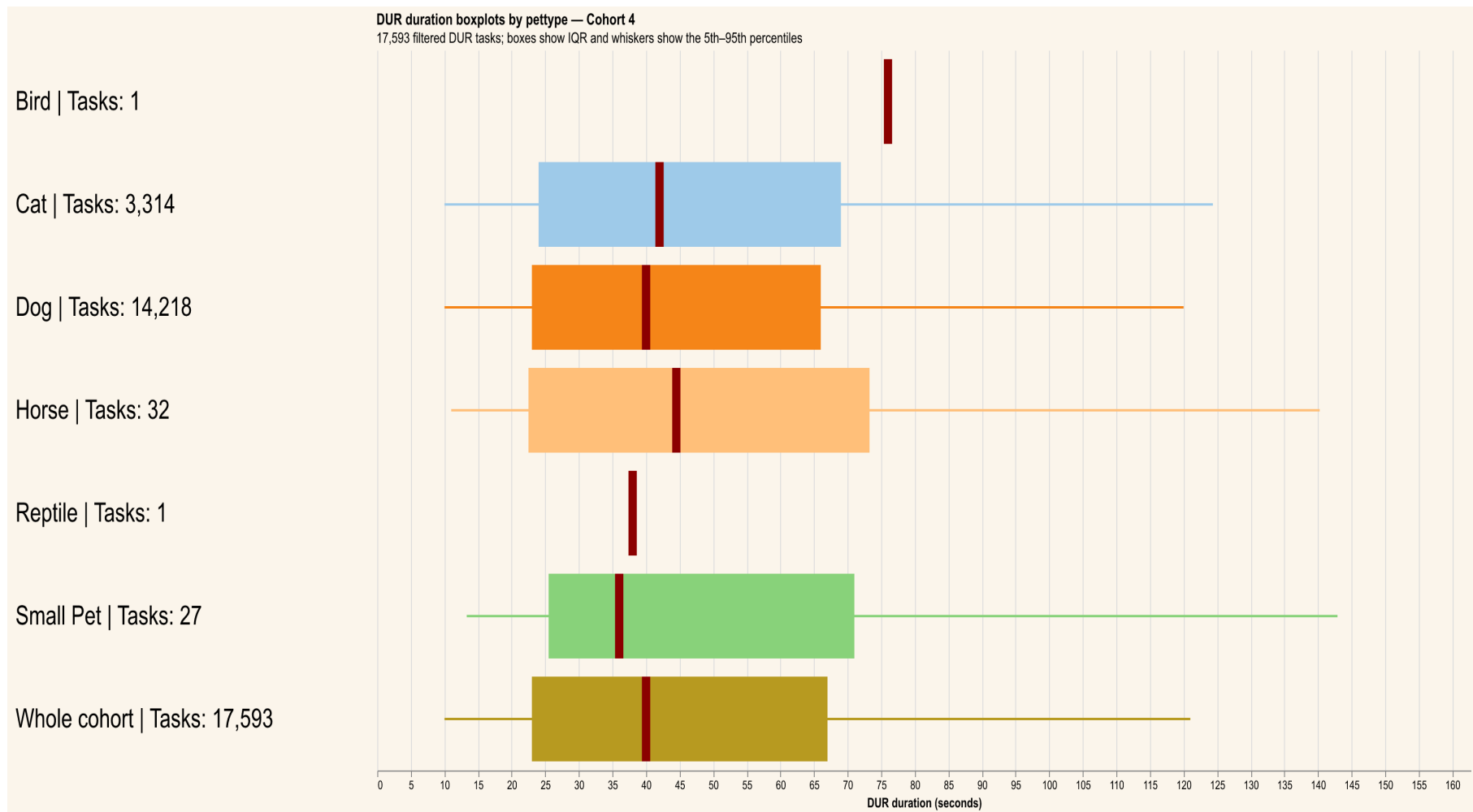


Pet Types Matter

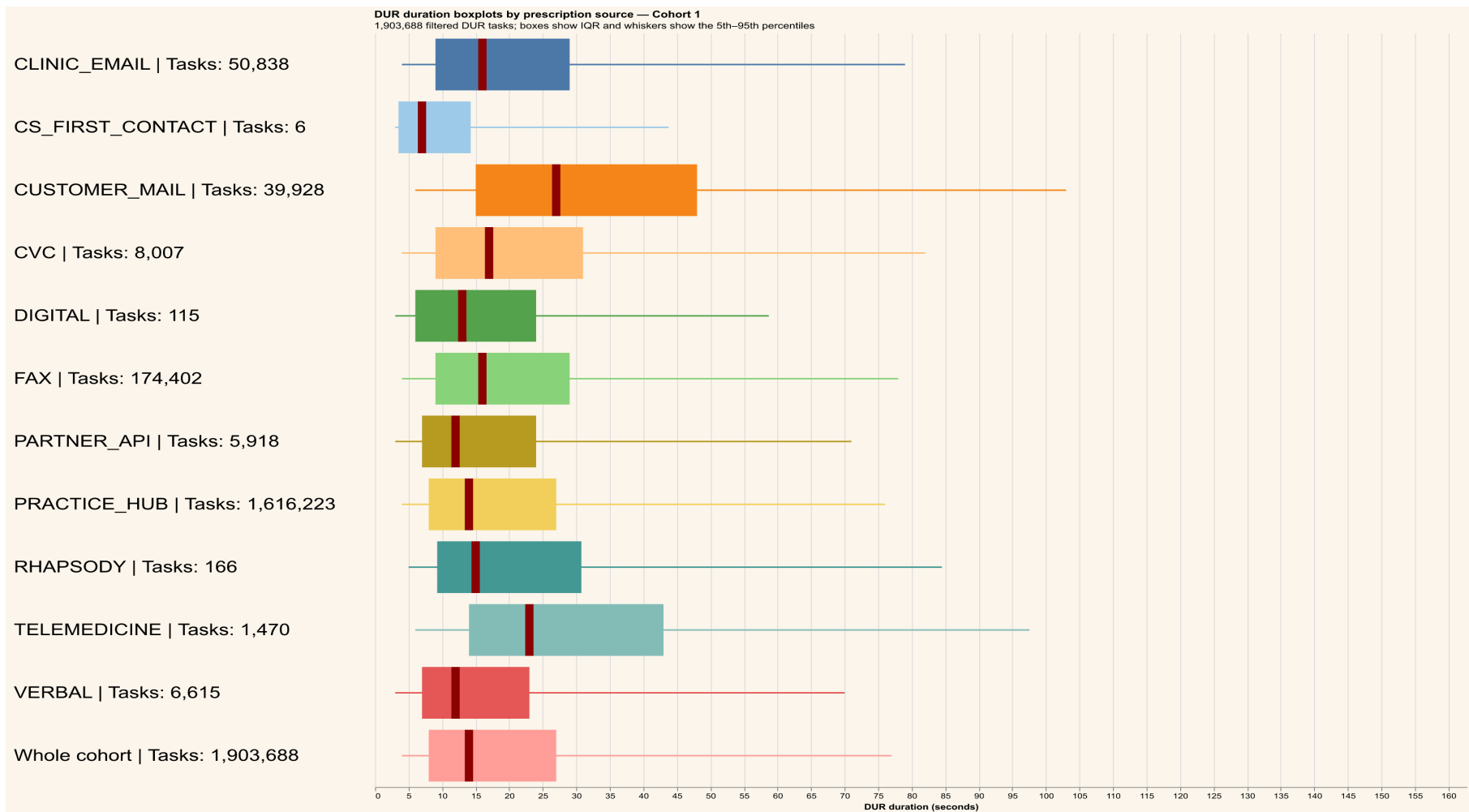


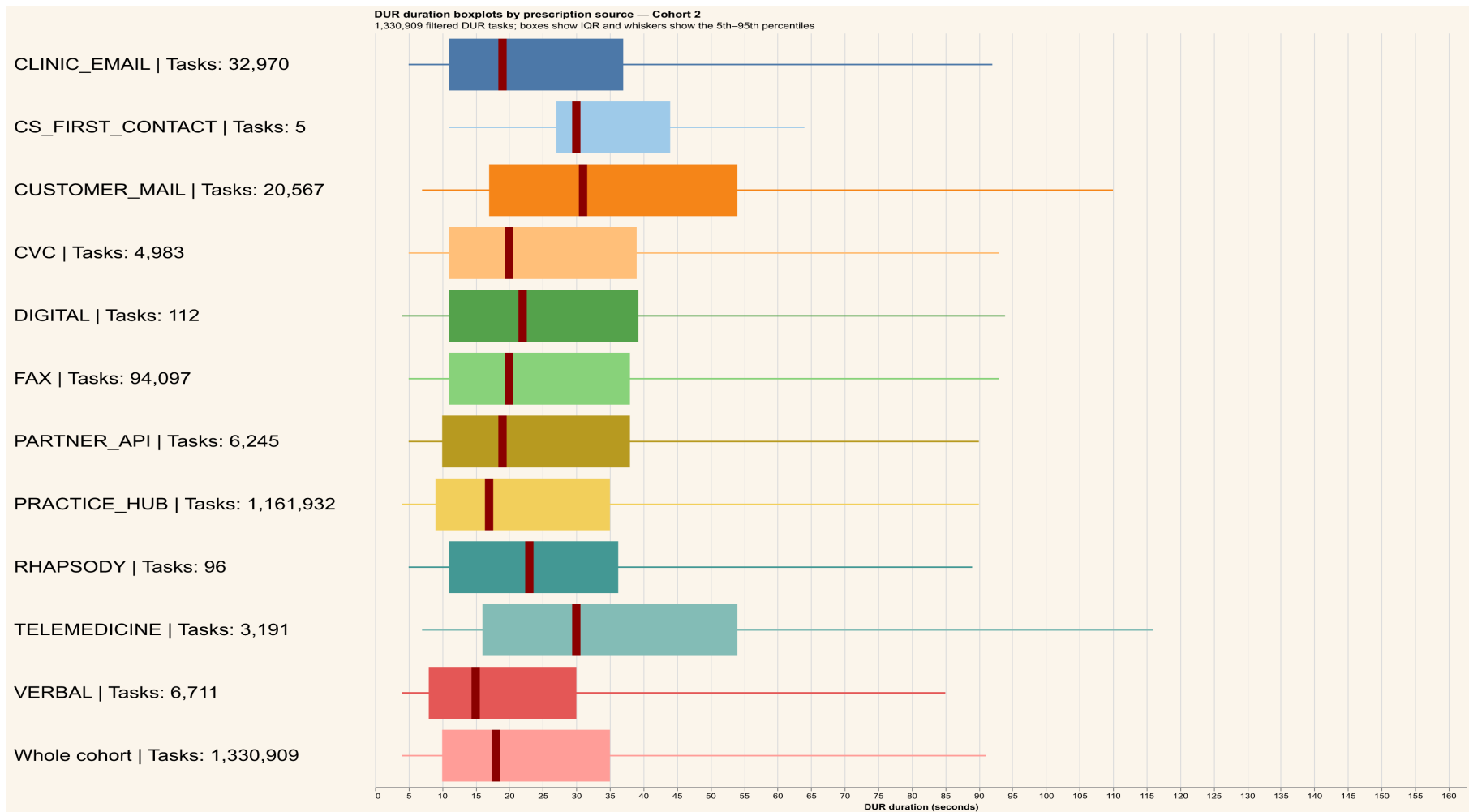


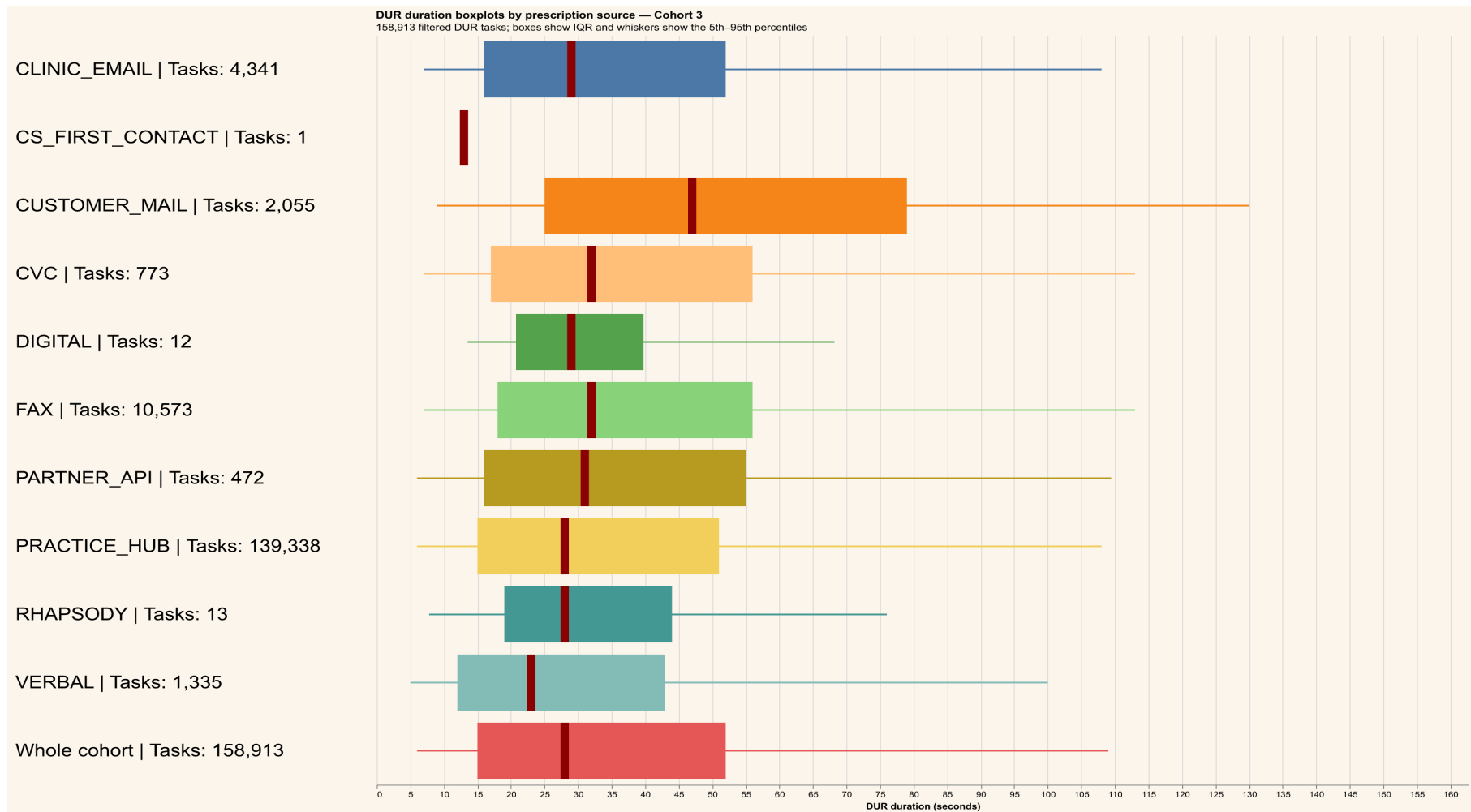


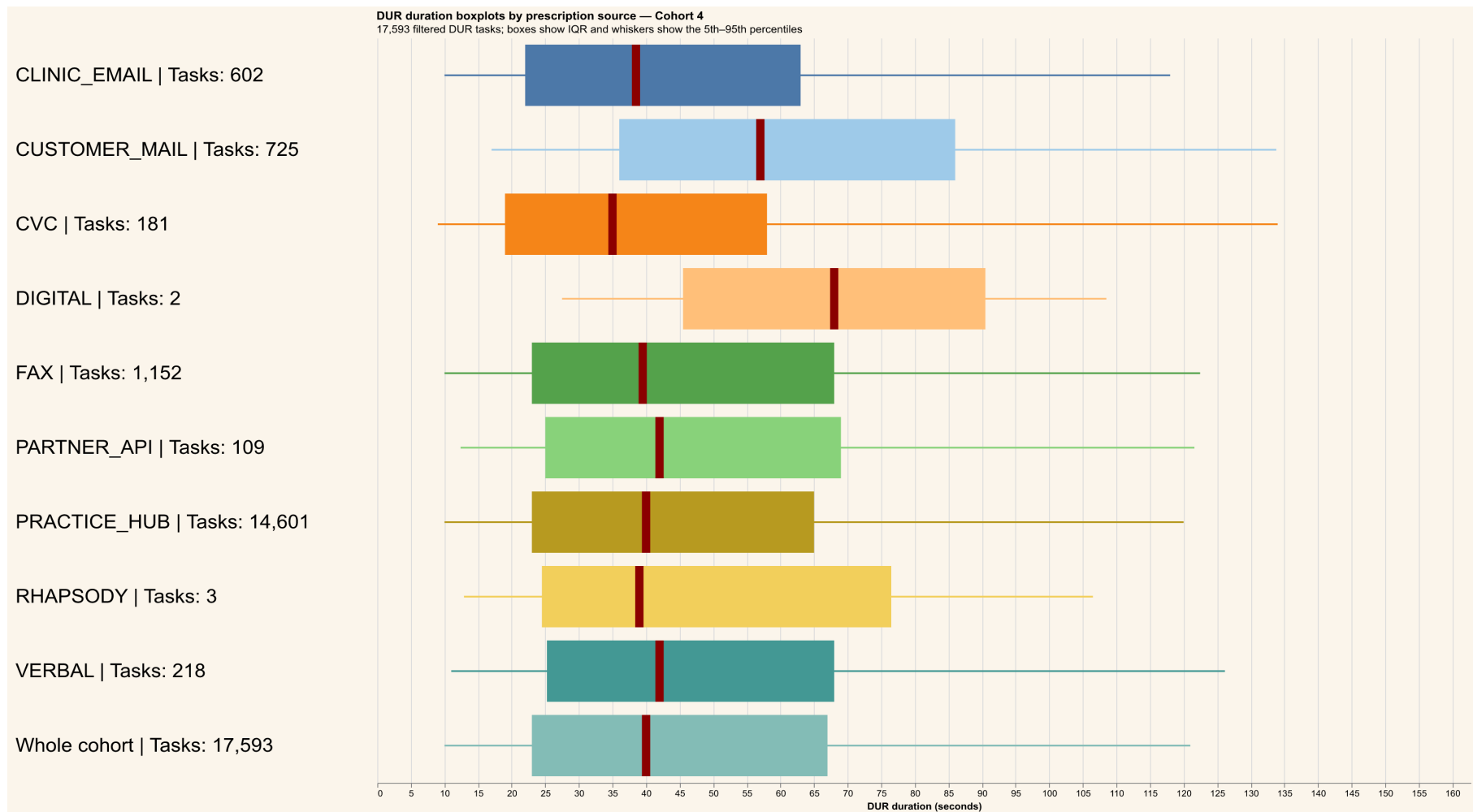


Prescription Source

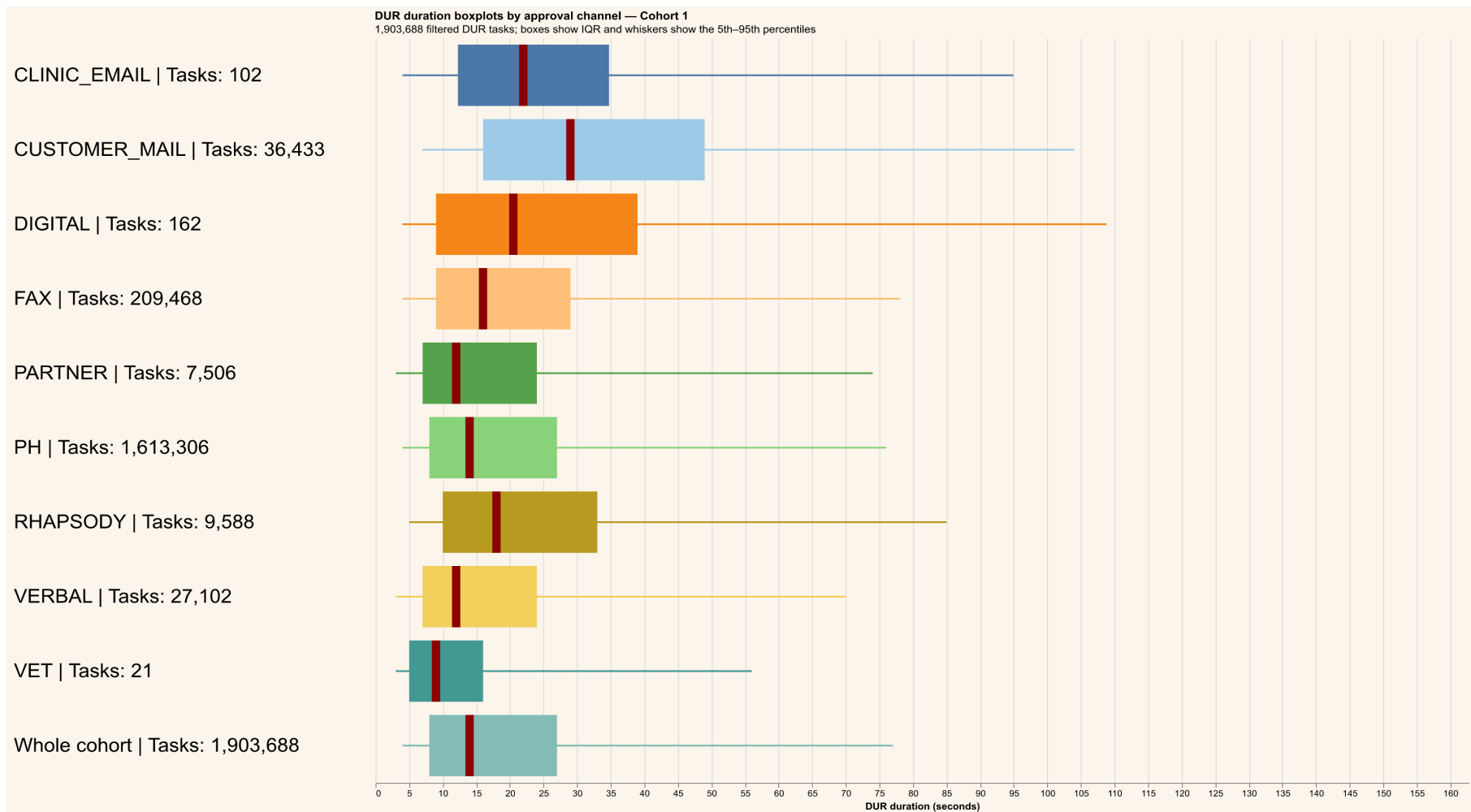


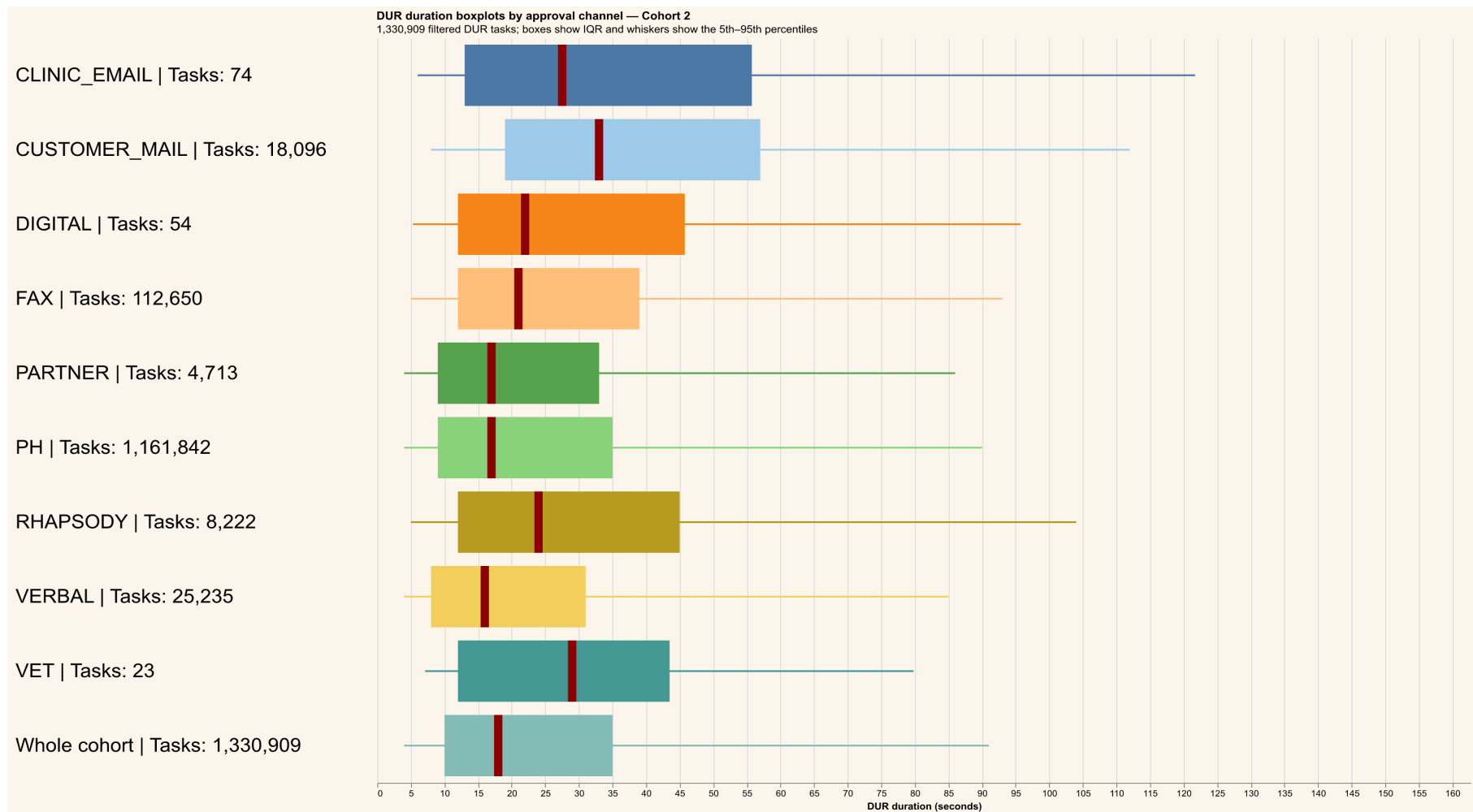


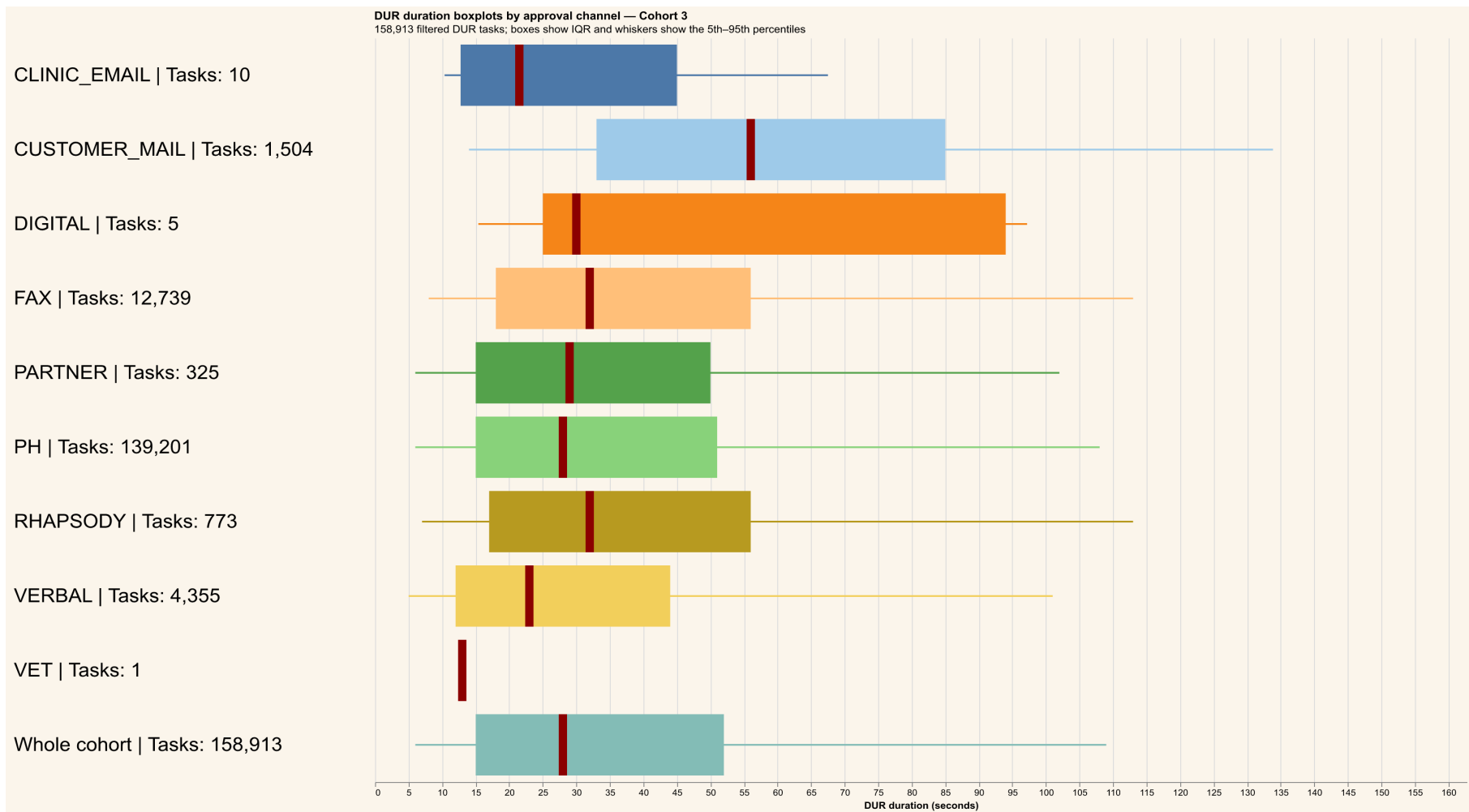


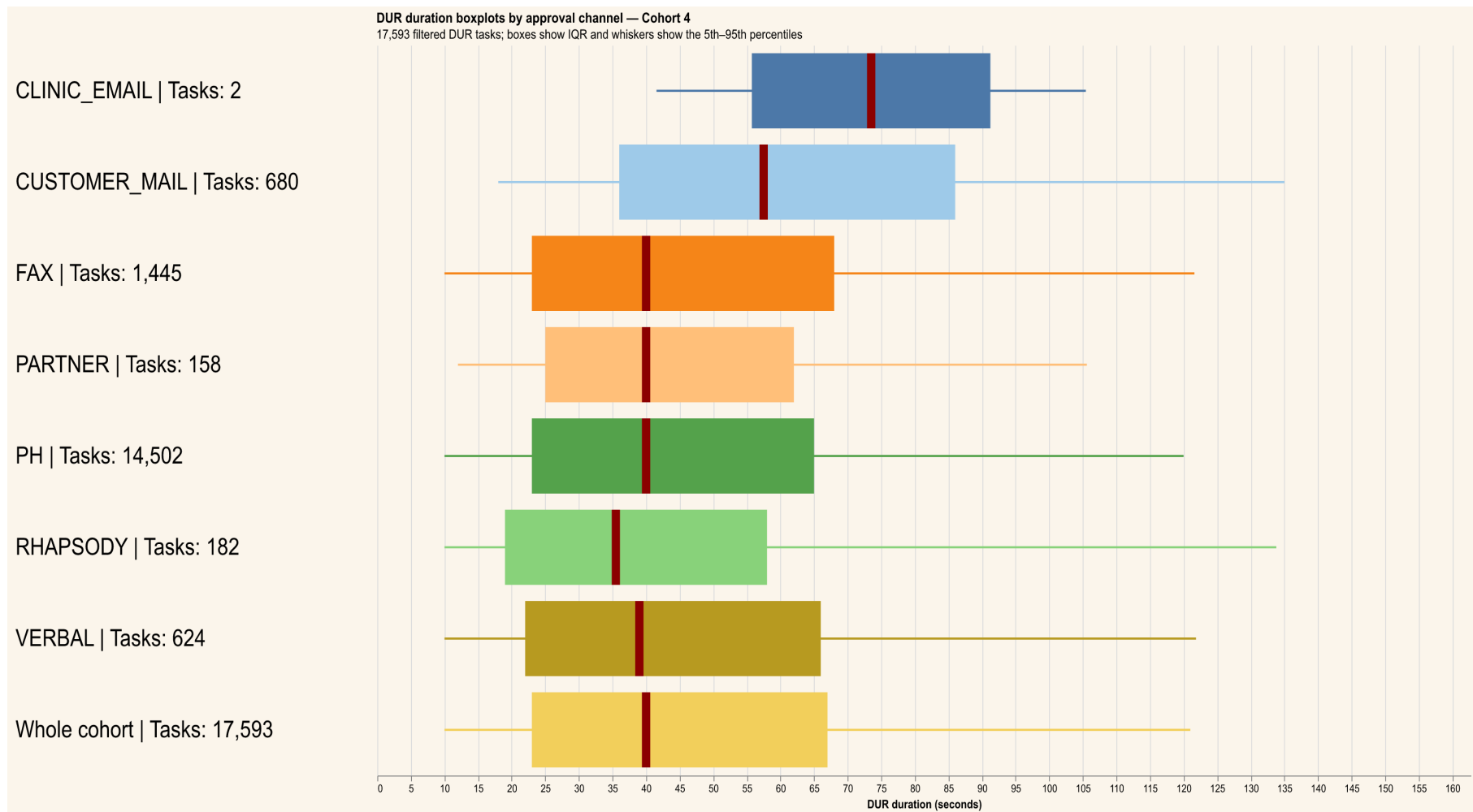


Approval Channel









Cohort Descriptions

- Cohort 1: Boxed parasiticides. This was the best starting point because it represents roughly 40% of total DUR volume and the majority of the initial digital volume, while the clinical rules are relatively standardized and label-driven. The primary checks are species, weight range, monthly directions, days' supply, duplicate parasite coverage, and known contraindications. This gave us a high-volume population with relatively low variability for validating the core RxBuddy and DUR Copilot experience.
- Cohort 2: Non-controlled, non-boxed medications. This represents roughly 29% of total DUR volume and materially expands coverage, but the clinical review is more complex. These medications require generalized dosing evaluation, including strength, dose, frequency, route, indication, species appropriateness, duplicate therapy, contraindications, and potential interactions. We separated this cohort so those broader AI checks could be validated without also introducing compounding or controlled-substance requirements at the same time.
- Cohort 3: Non-controlled compounded medications. This is a smaller segment at roughly 4.4% of DUR volume, but it has distinct clinical and compliance

requirements. Compounds require a documented compounding reason, formulation-specific directions, and validation that the prescribed days' supply does not exceed the product's beyond-use date. BUD rules vary by dosage form, storage conditions, formulation, and stability-study results. Keeping compounds separate allowed us to test these specialized workflows and manage their impact on pharmacist throughput independently.

- Cohort 4: gabapentin and the high-scrutiny regulatory workflow. This represents roughly 1.1% of DUR volume. The implemented Phase 4 scope focused on uncontrolled gabapentin rather than the full controlled-substance population, but it was separated because it introduces site eligibility, pharmacist skill gating, regulatory indicators, mixed-order handling, and state-specific scrutiny. It also gave us a contained way to validate infrastructure that will be needed for broader controlled-substance support later.